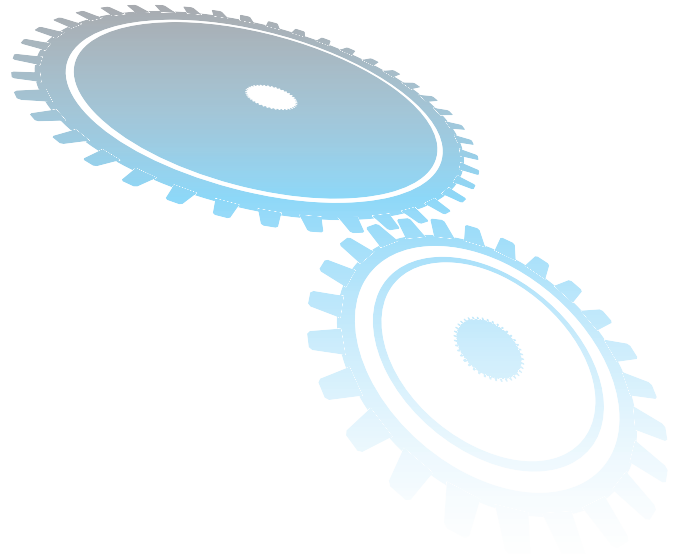


7

Spur Gears



Gears are machine elements used to transmit rotary motion between two shafts, usually with a constant speed ratio. In this chapter, we will discuss the case where the axes of the two shafts are parallel, and the teeth are straight and parallel to the axes of rotation of the shafts; such gears are called *spur gears*.

7.1 TERMINOLOGY AND DEFINITIONS

A pair of spur gears in mesh is shown in Fig. 7.1. The *pinion* is a name given to the smaller of the two mating gears; the larger is often called the *gear* or the *wheel*. The pair of gears, chosen to work together, is often called a *gearset*.

The terminology of gear teeth is shown in Fig. 7.2, where most of the following definitions are shown.

The *pitch circle* is a theoretical circle on which all calculations are based. The pitch circles of a pair of mating gears are tangent to each other, and it is these pitch circles that were pictured in earlier chapters as rolling against each other without slip.

The *diametral pitch* P is the ratio of the number of teeth on the gear to its pitch diameter; that is,

$$P = \frac{N}{2R}, \quad (7.1)$$

where N is the number of teeth, and R is the pitch circle radius. Note that the diametral pitch cannot be directly measured on the gear itself. Also, note that, as the value of the diametral pitch becomes larger, the teeth become smaller; this is shown clearly in Fig. 7.3. The diametral pitch is used to indicate the tooth size in US customary units and usually has units of teeth per inch. A pair of mating gears has the same diametral pitch.

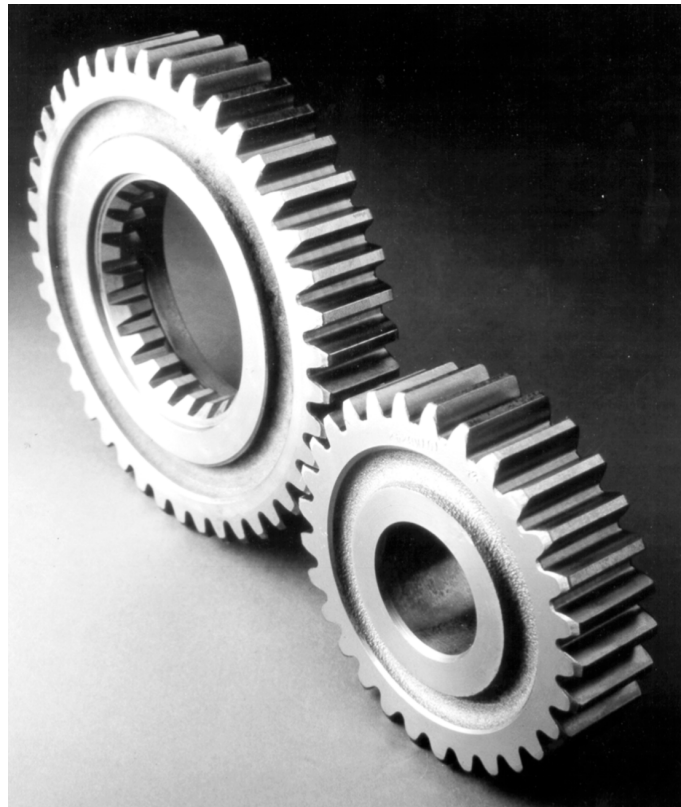


Figure 7.1 Pair of spur gears in mesh. (Courtesy of Gleason Works, Rochester, NY).

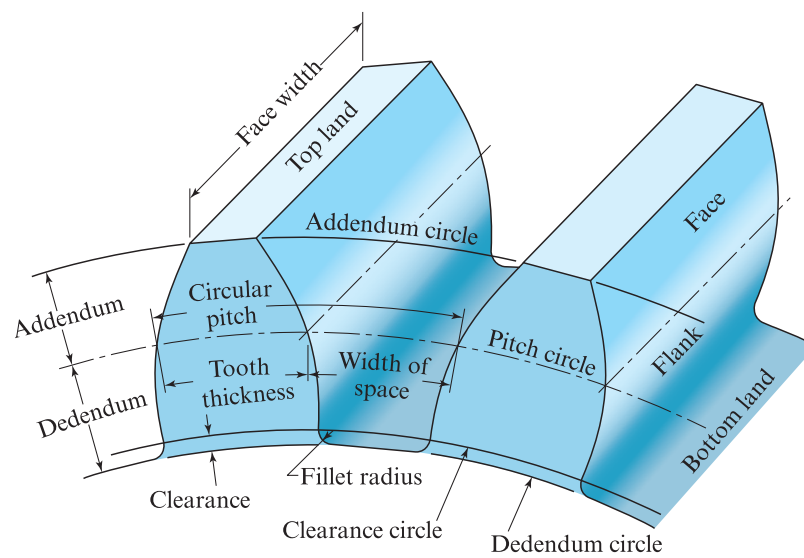


Figure 7.2 Gear tooth terminology.

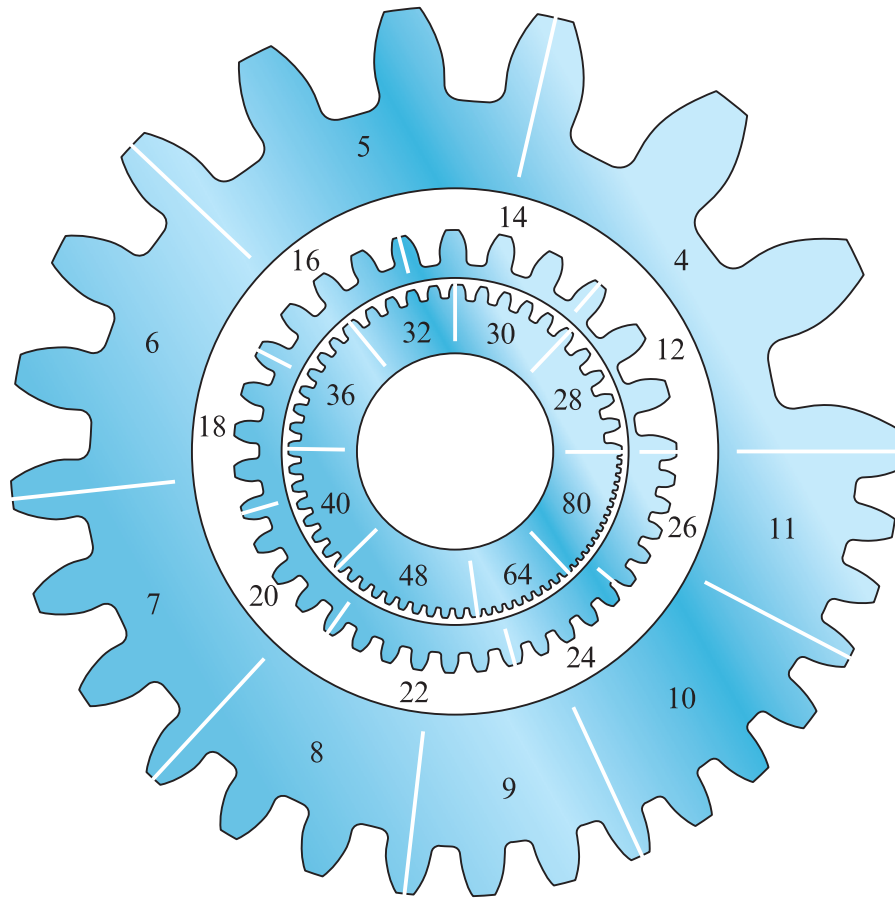


Figure 7.3 Tooth sizes in teeth per inch for various diametral pitches. (Courtesy of Gleason Cutting Tools Corp., Loves Park, IL.)

The *module* m is the ratio of the pitch diameter of the gear to its number of teeth; that is,

$$m = \frac{2R}{N}. \quad (7.2)$$

The module is the usual unit for indicating tooth size in International System (SI) units, and it customarily has units of millimeters per tooth. Note that the module is the reciprocal of the diametral pitch, and the relationship can be written as

$$m = \frac{25.4 \text{ (mm/in)}}{P \text{ (teeth/in)}} = \frac{25.4}{P} \text{ mm/tooth.}$$

Also note that metric gears should not be interchanged with US gears, since their standards for tooth sizes are not the same.

The *circular pitch*, p , is the distance from one tooth to the adjacent tooth, measured along the pitch circle. Therefore, it can be determined from

$$p = \frac{2\pi R}{N}. \quad (7.3)$$

Circular pitch is related to the previous definitions, depending on the units, by

$$p = \frac{\pi}{P} = \pi m. \quad (7.4)$$

The *addendum*, a , is the radial distance from the pitch circle to the top land of each tooth.

The *dedendum*, d , is the radial distance from the pitch circle to the bottom land of each tooth.

The *whole depth* is the sum of the addendum and dedendum.

The *clearance*, c , is the amount by which the dedendum of a gear exceeds the addendum of the mating gear.

The *backlash* is the amount by which the width of a tooth space exceeds the thickness of the engaging tooth measured along the pitch circles.

7.2 FUNDAMENTAL LAW OF TOOTHED GEARING

Gear teeth mating with each other to produce rotary motion are similar to a cam and follower. When the tooth profiles (or cam and follower profiles) are shaped to produce a constant angular velocity ratio between the two shafts, then the two mating surfaces are said to be *conjugate*. It is possible to specify an arbitrary profile for one tooth and then to find a profile for the mating tooth so that the two surfaces are conjugate. One possible choice for such conjugate profiles is the *involute* profile, which, with few exceptions, is in universal use for gear teeth.

A single pair of mating gear teeth must be shaped such that, as they pass through their entire period of contact, the ratio of the angular velocity of the driven gear to that of the driving gear—that is, the first-order kinematic coefficient—*must remain constant*. This is the fundamental criterion that governs the choice of the tooth profiles. If this were not true in gearing, very serious vibration and impact problems would result, even at low speeds.

In Sec. 3.17 we learned the angular-velocity-ratio theorem, which states that the first-order kinematic coefficient of any mechanism is inversely proportional to the segments into which the common instant center cuts the line of centers. In Fig. 7.4, two profiles are in contact at point T ; let profile 2 represent the driver and profile 3 represent the driven element. The normal to the surfaces, CD , is called the *line of action*. The normal to the profiles at the point of contact, T , intersects the line of centers O_2O_3 at the instant center of velocity. In gearing, this instant center is referred to as the *pitch point* and usually carries the label P .

Designating the pitch circle radii of the two gear profiles as R_2 and R_3 , from the angular-velocity-ratio theorem, Eq. (3.28), we see that

$$\frac{\omega_2}{\omega_3} = \frac{R_3}{R_2}. \quad (7.5)$$

This equation is frequently used to define the fundamental law of gearing, which states that as gears go through their mesh, *the pitch point must remain stationary on the line of*

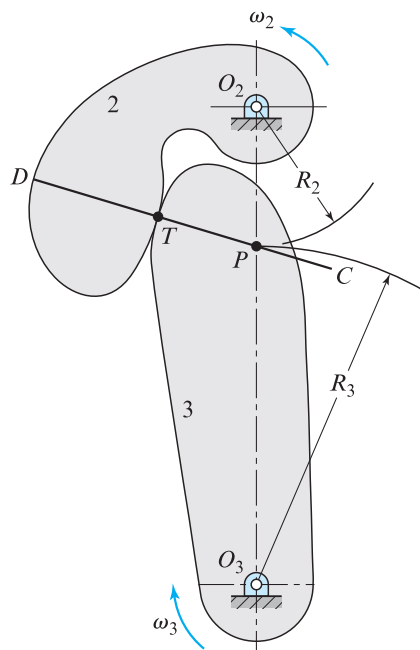


Figure 7.4

centers so that the speed ratio remains constant. This means that the line of action for every new instantaneous point of contact must always pass through the stationary pitch point, P . Thus, the problem of finding a conjugate profile for a given shape is to find a mating shape that satisfies the fundamental law of gearing.

It should not be assumed that any shape or profile is satisfactory just because a conjugate profile can be found. Although, theoretically, conjugate curves might be found, the practical problems of reproducing these curves from steel gear blanks or other materials while using existing machinery still exist. In addition, the sensitivity of the law of gearing to small dimensional changes of the shaft center distance caused either by misalignment or by large forces must also be considered. Finally, the tooth profile selected must be one that can be reproduced quickly and economically in very large quantities. A major portion of this chapter is devoted to illustrating how the involute curve profile fulfills these requirements.

7.3 INVOLUTE PROPERTIES

An *involute* curve is the path generated by a tracing point on a taut cord as the cord is unwrapped from a cylinder called the *base cylinder*. This is shown in Fig. 7.5, where T is the tracing point. Note that the cord, AT , is normal to the involute at T , and distance AT is the instantaneous value of the radius of curvature. As the involute is generated from its origin, T_0 , to T_1 , the radius of curvature varies continuously; it is zero at T_0 and increases continuously to T_1 . Thus, the cord is the generating line, and it is always normal to the involute.

If the two mating tooth profiles both have the shapes of involute curves, the condition that the pitch point, P , remain stationary is satisfied. This is shown in Fig. 7.6, where two gear blanks with fixed centers O_2 and O_3 are shown having base cylinders with respective

Figure 7.5 Involute curve.

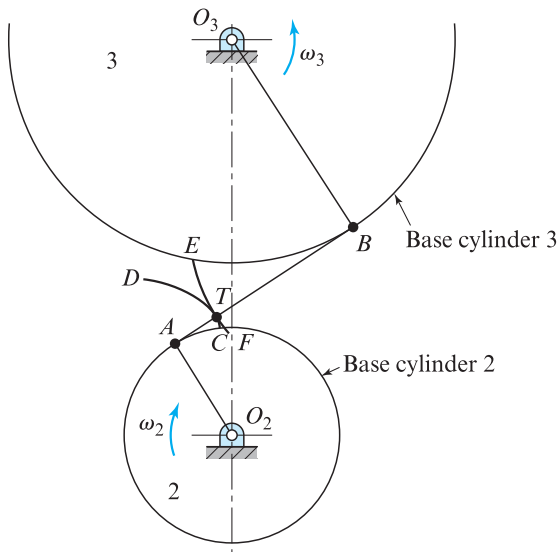
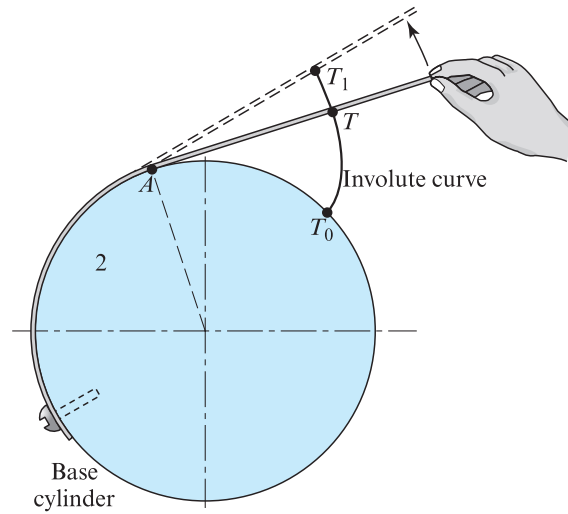


Figure 7.6 Conjugate involute curves.

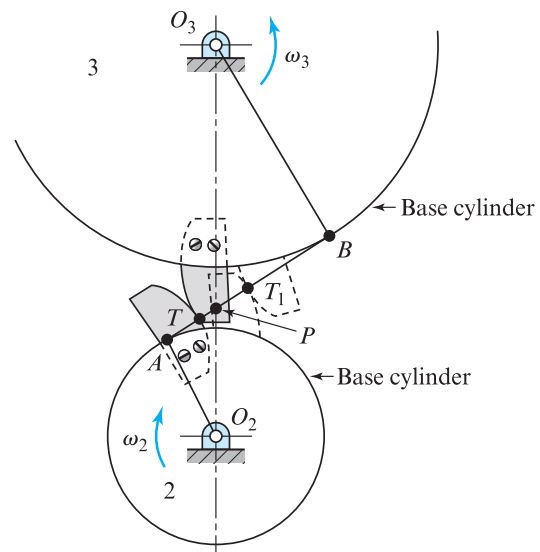


Figure 7.7 Involute action.

radii of O_2A and O_3B . We now imagine that a cord is wound clockwise around the base cylinder of gear 2, pulled taut between points A and B , and wound counterclockwise around the base cylinder of gear 3. If the two base cylinders are now rotated in opposite directions to keep the cord tight, a tracing point, T , traces out the involutes CD on gear 2 and EF on gear 3. The two involutes thus generated simultaneously by the single tracing point, T , are conjugate profiles.

Next imagine that the involutes of Fig. 7.6 are scribed on plates and that the plates are cut along the scribed curves and then bolted to the respective cylinders in the same postures. The result is shown in Fig. 7.7. The cord can now be removed, and, if gear 2 is moved clockwise, gear 3 is caused to move counterclockwise by the camlike action of the

two curved plates. The path of contact is line AB , formerly occupied by the cord. Since line AB is the generating line for each involute, it is normal to both profiles at all points of contact. Also, it always occupies the same position since it is always tangent to both base cylinders. Therefore point P is the pitch point. Point P does not move; therefore, the involute curves are conjugate curves and satisfy the fundamental law of gearing.

7.4 INTERCHANGEABLE GEARS; AGMA STANDARDS

A *tooth system* is the name given to a *standard** that specifies the relationships among addendum, dedendum, clearance, tooth thickness, and fillet radius to attain interchangeability of gears of different tooth numbers but of the same pressure angle and the same diametral pitch or module. We should be aware of the advantages and disadvantages of such a tooth system so that we can choose the best gears for a given design and have a basis for comparison if we depart from a standard tooth profile.

For a pair of spur gears to properly mesh, they must share the same pressure angle and the same tooth size, as specified by the diametral pitch or module. The numbers of teeth and the pitch diameters of the two gears in mesh need not match but are chosen to give the desired speed ratio, as demonstrated in Eq. (7.5).

The sizes of the teeth used are chosen by selecting the diametral pitch, P , or module, m . Standard cutters are generally available for the sizes listed in Table 7.1. Once the diametral pitch or module is chosen, the remaining dimensions of the tooth are set by the standards in Table 7.2. Tables 7.1 and 7.2 contain the standards for the spur gears most in use today, and they include values for both SI and US customary units.

Table 7.1 Standard Gear Tooth Sizes

Standard diametral pitches P , US customary, teeth/in	
Coarse	1, 1 $\frac{1}{4}$, 1 $\frac{1}{2}$, 1 $\frac{3}{4}$, 2, 2 $\frac{1}{2}$, 3, 4, 5, 6, 8, 10, 12, 14, 16, 18
Fine	20, 24, 32, 40, 48, 64, 72, 80, 96, 120, 150, 200
Standard modules m , SI, mm/tooth	
Preferred	1, 1.25, 1.5, 2, 2.5, 3, 4, 5, 6, 8, 10, 12, 16, 20, 25, 32, 40, 50
Next choice	1.125, 1.375, 1.75, 2.25, 2.75, 3.5, 4.5, 5.5, 7, 9, 11, 14, 18, 22, 28, 36, 45

* Standards are defined by the American Gear Manufacturers Association (AGMA) and the American National Standards Institute (ANSI). The AGMA standards may be quoted or extracted in their entirety, provided that an appropriate credit line is included—for example, “Extracted from AGMA Information Sheet—Strength of Spur, Helical, Herringbone, and Bevel Gear Teeth (AGMA 225.01) with permission of the publisher, the American Gear Manufacturers Association, 1500 King Street, Suite 201, Alexandria, VA 22314.” These standards have been used extensively in Chaps. 7 and 8.

Table 7.2 Standard Tooth Systems for Spur Gears

System	Pressure angle, ϕ (deg)	Addendum, a	Dedendum, d
Full depth	20°	$1/P$ or $1m$	$1.25/P$ or $1.25m$
Full depth	$22\frac{1}{2}^\circ$	$1/P$ or $1m$	$1.25/P$ or $1.25m$
Full depth	25°	$1/P$ or $1m$	$1.25/P$ or $1.25m$
Stub teeth	20°	$0.8/P$ or $0.8m$	$1/P$ or $1m$

Let us illustrate the design choices by an example.

EXAMPLE 7.1

Two parallel shafts, separated by a distance (commonly referred to as the *center distance*) of 3.5 in, are to be connected by a gear pair so that the output shaft rotates at 40% of the speed of the input shaft. Design a gearset to fit this situation.

SOLUTION

The center distance can be written as $R_2 + R_3 = 3.5$ in, and substituting the given information into Eq. (7.5), we have $\omega_3/\omega_2 = R_2/R_3 = 0.40$. Then, substituting the second equation into the first equation and rearranging, we find that $R_2 = 1.0$ in and $R_3 = 2.5$ in. Next, we must choose the size of the teeth by picking a value for the diametral pitch or module. From Eq. (7.1), we find the number of teeth on the two gears to be $N_2 = 2PR_2 = 2P$ and $N_3 = 2PR_3 = 5P$. The choice of P or m for tooth size is often iterative. First, we might choose a value of $P = 6$ teeth/in; this gives the numbers of teeth as $N_2 = 12$ teeth and $N_3 = 30$ teeth; if we choose $P = 10$ teeth/in, then we get $N_2 = 20$ teeth and $N_3 = 50$ teeth. At this time, either choice appears acceptable, and we choose $P = 10$ teeth/in. However, this choice of P (or m) must later be checked for possible undercutting, as we will study in Sec. 7.7; for contact ratio, which we will study in Sec. 7.8; and for strength and wear of the teeth [4].

7.5 FUNDAMENTALS OF GEAR-TOOTH ACTION

To illustrate the fundamentals, we now proceed, step by step, through the actual graphic layout of a pair of spur gears. The dimensions used are those of Example 7.1, assuming standard 20° full-depth involute tooth form as specified in Table 7.2. The various steps, in the correct order, are shown in Figs. 7.8 and 7.9 and are as follows.

STEP 1 Calculate the two pitch circle radii, R_2 and R_3 , as in Example 7.1; identify O_2 and O_3 as the two shaft centers (Fig. 7.8); and draw the two pitch circles tangent to each other.

STEP 2 Draw the common tangent to the pitch circles perpendicular to the line of centers and through the pitch point P (Fig. 7.8). Draw the *line of action* at an angle equal to the pressure angle $\phi = 20^\circ$ from the common tangent. This line of action corresponds to the generating line discussed in Sec. 7.3; it is always normal to the involute curves and always passes through the pitch point. It is called the *line of action*, since the point of contact

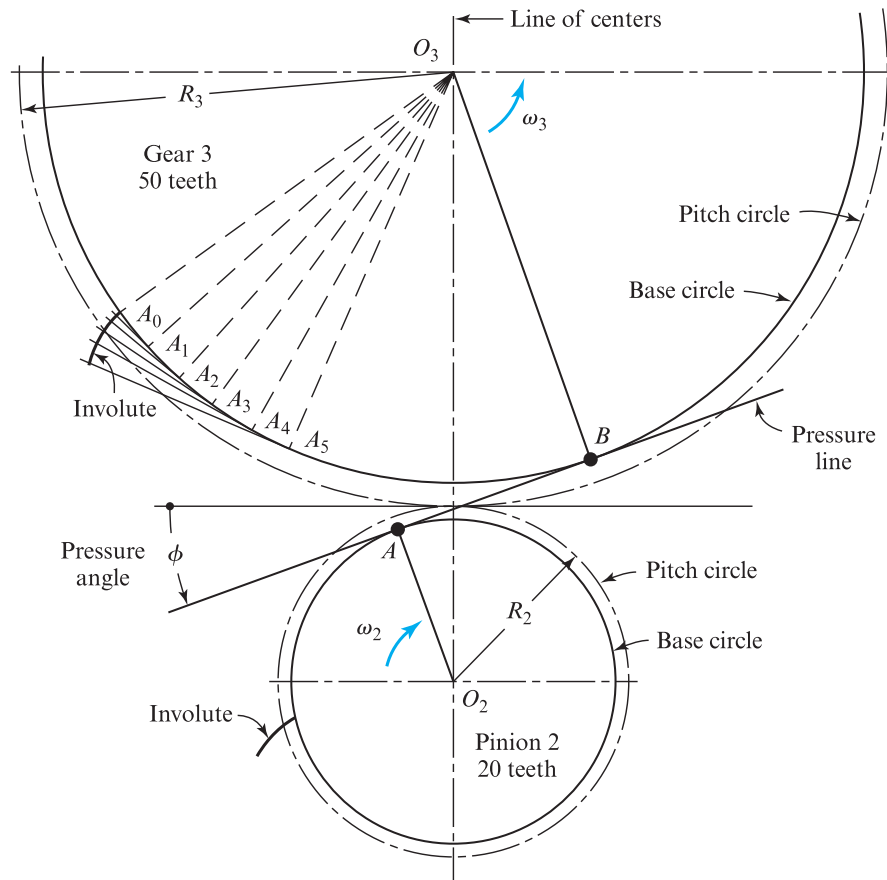


Figure 7.8 Start of layout of a pair of spur gears.

between the gear teeth always lies on this line, the pressure line, since, assuming no friction, the resultant tooth force acts along this line.

STEP 3 Through the centers of the two gears, draw the two perpendiculars, O_2A and O_3B , to the line of action (Fig. 7.8). Draw the two *base circles* with radii of $r_2 = O_2A$ and $r_3 = O_3B$; these correspond to the base cylinders of Sec. 7.3.

STEP 4 From Table 7.2, with $P = 10$ teeth/in, the addendum for both of the gears is found to be

$$a = \frac{1}{P} = \frac{1}{10 \text{ teeth/in}} = 0.10 \text{ in.}$$

Adding this to each of the pitch circle radii, draw the two addendum circles that define the top lands of the teeth on each gear. Carefully identify and label point C where the addendum circle of gear 3 intersects the line of action (Fig. 7.9). Similarly, identify and label point D where the addendum circle of gear 2 intersects the line of action.

Visualizing the rotation of the two gears in the directions given, we see that contact is not possible before point C , since the teeth of gear 3 are not of sufficient height; thus, C is the first point of contact between the teeth. Similarly, the teeth of gear 2 are too short to

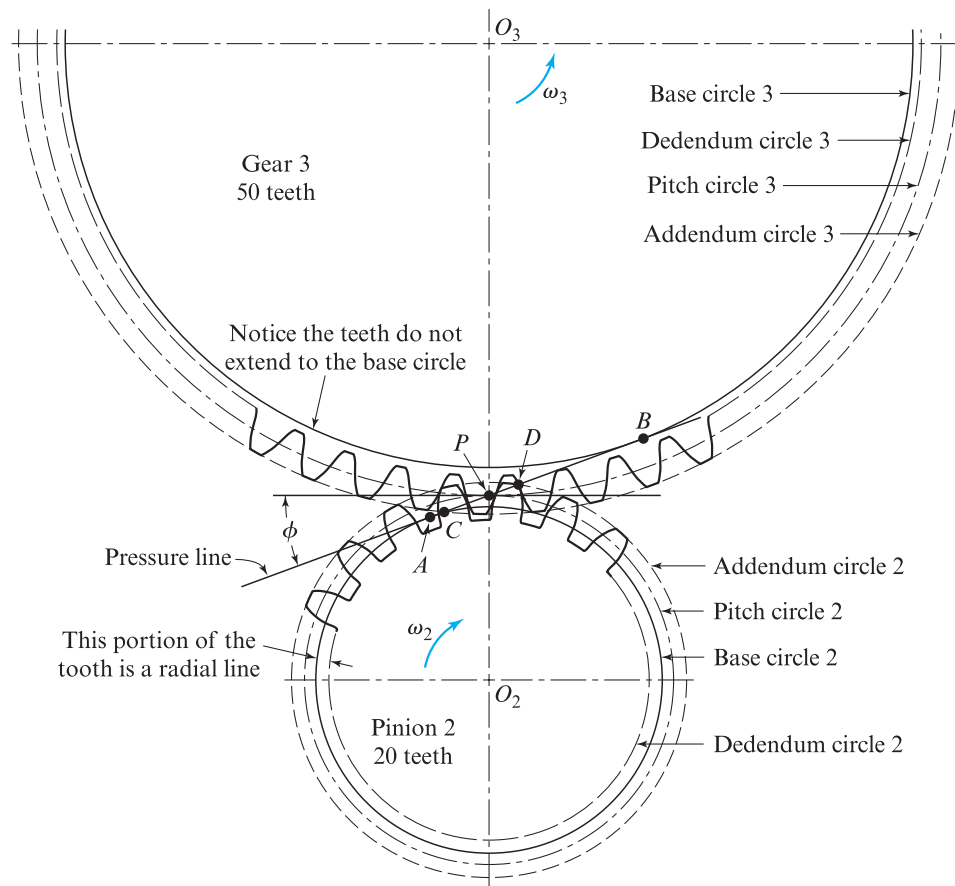


Figure 7.9 Continued layout of the pair of spur gears.

allow further contact after reaching point *D*; thus, contact between a pair of mating teeth begins at *C*, continues to *D*, and then ceases. When contact of a pair of teeth ceases at *D*, one or more pairs of trailing teeth must still be in contact in the span *CD* and take on the sharing of the transmitted load.

Steps 1 through 4 are critical for verifying the choice of any gear pair. We will continue with the diagram shown in Fig. 7.9 when we check for interference, undercutting, and contact ratio in later sections. However, to complete our visualization of gear tooth action, let us first proceed to the construction of the complete involute tooth shapes as shown in Fig. 7.8.

STEP 5 From Table 7.2, the dedendum for each gear is found to be

$$d = \frac{1.25}{P} = \frac{1.25}{10 \text{ teeth/in}} = 0.125 \text{ in.}$$

Subtracting this from each of the pitch circle radii, draw the two dedendum circles that define the bottom lands of the teeth on each gear (Fig. 7.9). Note that the dedendum circles often lie quite close to the base circles; however, they have distinctly different meanings. In

this example, the dedendum circle of gear 3 is larger than its base circle, and the dedendum circle of pinion 2 is smaller than its base circle. However, this is not always the case.

STEP 6 Generate an involute curve on each base circle as shown for gear 3 in Fig. 7.8. This is done by first dividing a portion of the base circle into a series of equal small parts, A_0, A_1, A_2 , and so on. Next, the radial lines O_3A_0, O_3A_1, O_3A_2 , and so on, are constructed, and tangents to the base circle are drawn perpendicular to each of these. The involute begins at A_0 . The second point is obtained by striking an arc, with center A_1 and radius A_0A_1 , up to the tangent line through A_1 . The next point is found by striking a similar arc with center at A_2 , and so on. This construction is continued until the involute curve is generated far enough to meet the addendum circle of gear 3. If the dedendum circle lies inside of the base circle, as is true for pinion 2 of this example, then, except for the fillet, the involute curve is extended inward to the dedendum circle by a radial line; this portion of the curve is not involute.

STEP 7 Using cardboard or preferably a sheet of clear plastic, cut a template for the involute curve and mark on it the center point of the corresponding gear. Note that two templates are needed, since the involute curves are different for gears 2 and 3.

STEP 8 Calculate the circular pitch using Eq. (7.4):

$$p = \frac{\pi}{P} = \frac{\pi}{10 \text{ teeth/in}} = 0.314 \text{ 16 in/tooth.}$$

This distance from one tooth to the next is now marked along the pitch circle, and the template is used to draw the involute portion of each tooth (Fig. 7.9). The width of a tooth and that of a tooth space are each equal to half of the circular pitch or $(0.314 \text{ 16 in/tooth})/2 = 0.157 \text{ 08 in/tooth}$. These distances are marked along the pitch circle, and the same template is turned over and used to draw the opposite sides of the teeth. The portion of the tooth space between the clearance and the dedendum may be used for a fillet radius. The top and bottom lands are now drawn as circular arcs along the addendum and dedendum circles to complete the tooth profiles. The same process is performed on the other gear using the other template.

Remember that steps 5 through 8 are not necessary for the proper design of a gearset. They are only included here to help us to visualize the relation between real tooth shapes and the theoretical properties of the involute curve.

Involute Rack We may imagine a *rack* as a spur gear having an infinitely large pitch diameter. Therefore, in theory, a rack is infinitely long and has an infinite number of teeth, and its base circle is located an infinite distance from the pitch point. For involute teeth, the curves on the sides of the teeth of a rack become straight lines making an angle with the line of centers equal to the pressure angle. The addendum and dedendum distances are the same as those given in Table 7.2. Figure 7.10 shows an involute rack in mesh with the pinion of the previous example.

Base Pitch Corresponding sides of involute teeth are parallel curves. The *base pitch* is the constant and fundamental distance between these curves—that is, the distance from one tooth to the next—measured along the common normal to the tooth profiles, which is the line of action (Fig. 7.10). The base pitch, p_b , and the circular pitch, p , are related as follows:

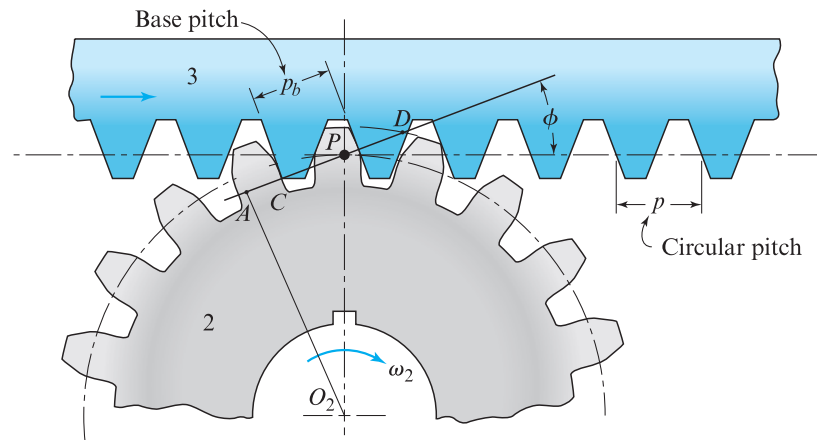


Figure 7.10 Involute pinion and rack.

$$p_b = p \cos \phi. \quad (7.6)$$

The base pitch is a much more fundamental measurement, as we will see later.

Internal Gear Figure 7.11 depicts the pinion of the preceding example in mesh with an *internal*, or *annular*, gear. With internal contact, both centers are on the same side of the pitch point. Thus, the locations of the addendum and dedendum circles of an internal gear are reversed with respect to the pitch circle; the addendum circle of the internal gear lies *inside* the pitch circle, whereas the dedendum circle lies *outside* the pitch circle. The base circle lies inside the pitch circle as with an external gear, but is now near the addendum circle. Otherwise, Fig. 7.11 is constructed in the same manner as was Fig. 7.9.

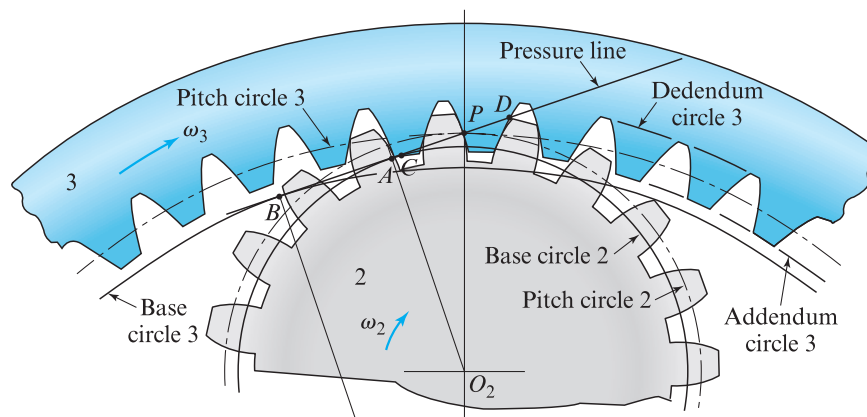


Figure 7.11 Involute pinion and internal gear.

7.6 MANUFACTURE OF GEAR TEETH

There are many ways of manufacturing the teeth of gears; for example, they can be made by sand casting, shell molding, investment casting, permanent-mold casting, die casting, or centrifugal casting. They can be formed by the powder-metallurgy process, or a single bar of aluminum can be formed by extrusion and then sliced into gears. Gears that carry large loads in comparison with their size are usually made of steel and are cut with either *form cutters* or *generating cutters*. In form cutting, the cutter is of the exact shape of the tooth space. With generating cutters, a tool having a shape different from the tooth space is moved through several cuts relative to the gear blank to obtain the proper shape for the teeth.

Probably the oldest method of cutting gear teeth is *milling*. A form milling cutter corresponding to the shape of the tooth space, such as that shown in Fig. 7.12*a*, is used to machine one tooth space at a time, as shown in Fig. 7.12*b*, after which the gear is indexed through one circular pitch to the next posture. Theoretically, with this method, a different cutter is required for each gear to be cut, since, for example, the shape of the tooth space in a 25-tooth gear is different from the shape of the tooth space in, say, a 24-tooth gear. Actually, the change in tooth space shape is not very large, and eight form cutters can be used to cut any gear in the range from 12 teeth to a rack with reasonable accuracy. Of course, a separate set of form cutters is required for each pitch.

Shaping is a highly favored method of *generating* gear teeth. The cutting tool may be either a rack cutter or a pinion cutter. The operation is explained by reference to Fig. 7.13. For shaping, the reciprocating cutter is first fed into the gear blank until the pitch circles are tangent. Then, after each cutting stroke, the gear blank and the cutter roll slightly on their pitch circles. When the blank and cutter have rolled by a distance equal to the circular pitch, one tooth has been generated and the cutting continues with the next tooth until all teeth have been cut. Shaping of an internal gear with a pinion cutter is shown in Fig. 7.14.

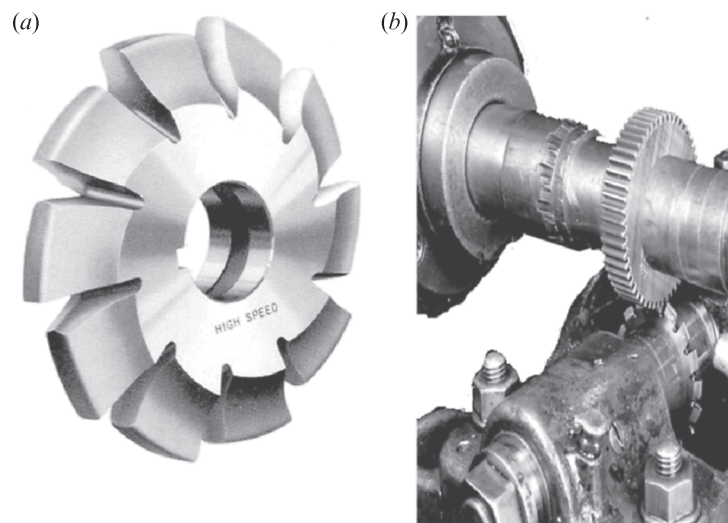
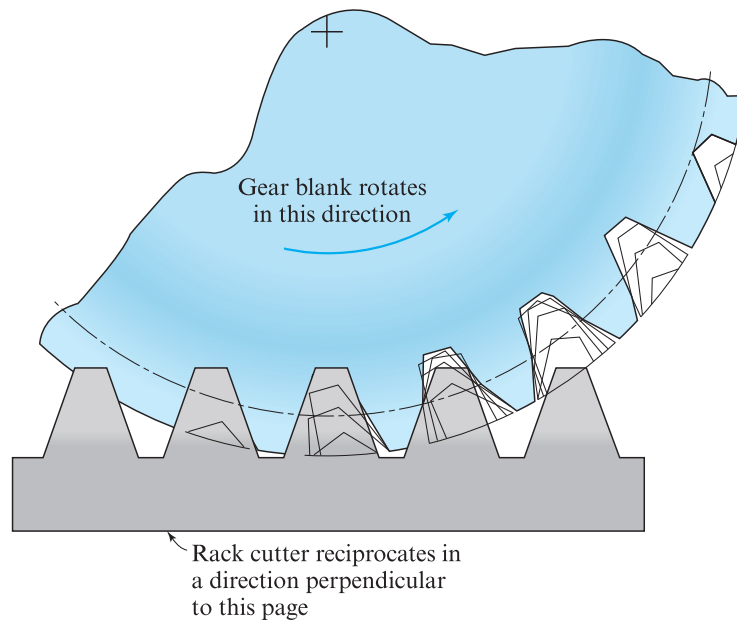


Figure 7.12 Manufacture of gear teeth by a form cutter: (a) a single-tooth involute hob; (b) machining of a single tooth space. (Courtesy of Gleason Works, Rochester, NY.)



 **Figure 7.13** Shaping of involute teeth with a rack cutter.

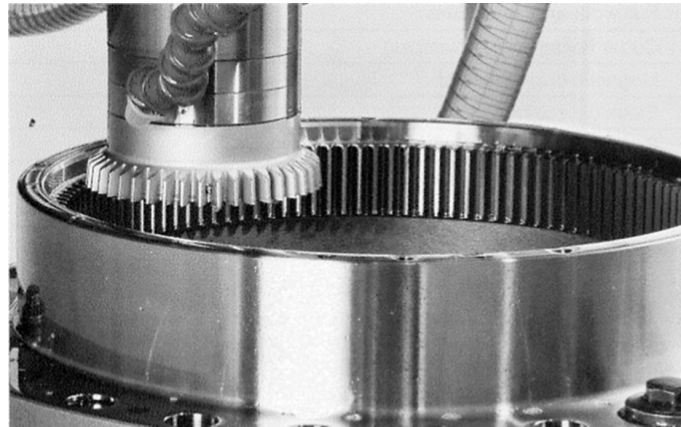


Figure 7.14 Shaping of an internal gear with a pinion cutter. (Courtesy of Gleason Works, Rochester, NY.)

Hobbing is another method of generating gear teeth, which is quite similar to shaping them with a rack cutter. However, hobbing is done with a special tool called a *hob*, a cylindric cutter with one or more helical threads quite like a screw-thread tap; the threads have straight sides like a rack. A number of different gear hobs are displayed in Fig. 7.15. A view of the hobbing of a gear is shown in Fig. 7.16. The hob and the gear blank are both



Figure 7.15 A variety of involute gear hobs. (Courtesy of Gleason Works, Rochester, NY.)



Figure 7.16 The hobbing of a gear. (Courtesy of Gleason Works, Rochester, NY.)

rotated continuously at the proper angular velocity ratio, and the hob is fed slowly across the face of the blank to cut the full thickness of the teeth.

Following the cutting process, grinding, lapping, shaving, and burnishing are often used as final finishing processes when tooth profiles of very high accuracy and surface finish are desired.

7.7 INTERFERENCE AND UNDERCUTTING

Figure 7.17 shows the pitch circles of the same gears used for discussion in Sec. 7.5. Let us assume that the pinion is the driver and that it is rotating clockwise.

We saw in Sec. 7.5 that for involute teeth, contact always takes place along the line of action, AB . Contact begins at point C , where the addendum circle of the driven gear crosses the line of action. Thus, initial contact is on the tip of the driven gear tooth and on the flank of the pinion tooth.

As the pinion tooth drives the gear tooth, contact approaches the pitch point, P . Near the pitch point, contact slides *up* the flank of the pinion tooth and *down* the face of the gear tooth. At the pitch point, contact is at the pitch circles; note that P is the instant center of velocity, and therefore the motion must be rolling with no slip at that point. Note also that this is the only location where the motion is true rolling.

As the teeth recede from the pitch point, the point of contact continues to travel in the same direction as before along the line of action. Contact continues to slide *up* the face of the pinion tooth and *down* the flank of the gear tooth. The last point of contact occurs at the tip of the pinion and the flank of the gear tooth, at the intersection, D , of the line of action and the addendum circle of the pinion.

The *approach* phase of the motion is the interval between the initial contact at point C and the pitch point, P . The *angles of approach* are the angles through which the two gears rotate as the point of contact progresses from C to P . However, reflecting on the unwrapping cord analogy of Fig. 7.6, we see that the distance CP is equal to a length

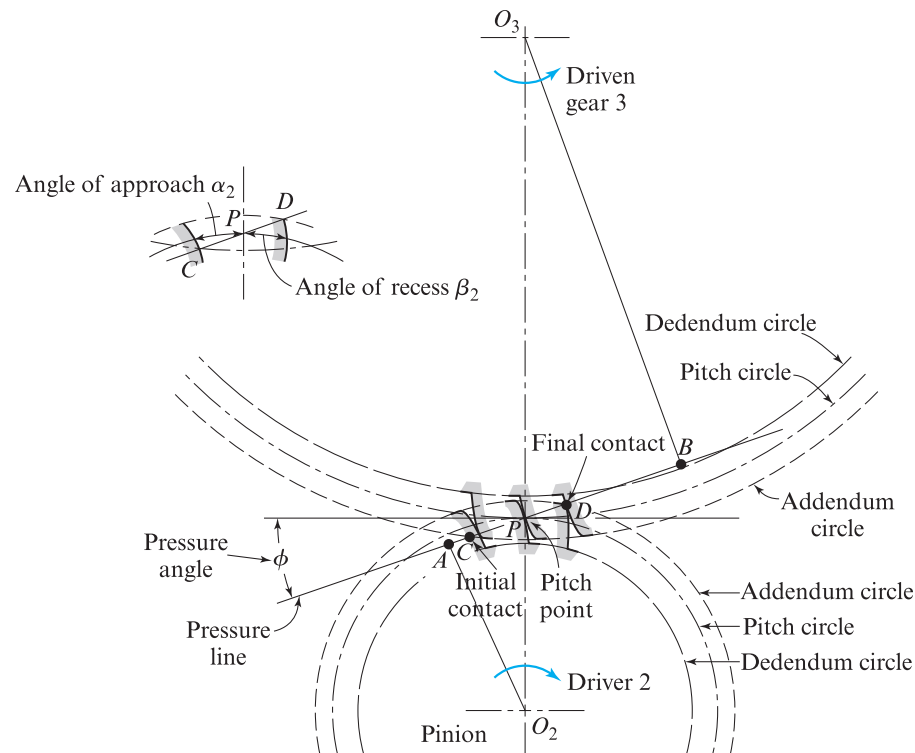


Figure 7.17 Approach and recess phases of gear tooth action.

$$\alpha_2 = \frac{CP}{r_2} \quad \text{and} \quad \alpha_3 = \frac{CP}{r_3}. \quad (7.7)$$
$$\beta_2 = \frac{PD}{r_2} \quad \text{and} \quad \beta_3 = \frac{PD}{r_3}. \quad (7.8)$$

Diagram illustrating interference in a spur gear pair. The diagram shows two gears, a driving gear (2) and a driven gear (3), with their respective pitch circles, base circles, and addendum circles. The pressure line is shown as a dashed line. The interference is highlighted in a blue shaded area on the flank of the driver gear (2) during approach. Labels indicate that portions of the profiles are not involute and that the interference occurs on the flank of the driver.

* Such gears were part of an older standard and are now obsolete. They are chosen here only to illustrate an example of interference.

full-depth involute teeth. The driver, gear 2, turns clockwise. As with previous figures, the points labeled *A* and *B* indicate the points of tangency of the line of action with the two base circles, whereas the points labeled *C* and *D* indicate the initial and final points of contact. Note that points *C* and *D* are now *outside* points *A* and *B*. This indicates interference.

The interference is explained as follows. Contact begins when the tip of the driven gear 3 tooth contacts the flank of the driving tooth. In this case, the flank of the driving tooth first tries to make contact with the driven tooth at point *C*, and this occurs *before* the involute portion of the driving tooth comes within range. In other words, contact occurs before the two teeth become tangent. The actual effect is that the non-tangent tip of the driven gear tooth interferes with and digs into the flank of the driving tooth.

In this example, a similar effect also occurs as the teeth leave contact. Contact should end at or before point *B*. Since, for this example, contact does not end until point *D*, the effect is for the non-tangent tip of the driving tooth to interfere with and dig into the flank of the driven tooth.

When gear teeth are produced by a generating process, interference is automatically eliminated, since the cutting tool removes the interfering portion of the tooth flank. This effect is called *undercutting*. If undercutting is at all pronounced, the undercut tooth is considerably weakened. Thus, the effect of eliminating interference by a generation process is merely to substitute another problem for the original.

The importance of the problem of teeth that have been weakened by undercutting cannot be emphasized too strongly. Of course, interference can be eliminated by using more teeth on the gears. However, if the gears are to transmit a given amount of power, more teeth can be used only by increasing the pitch diameter. This makes the gears larger, which is seldom desirable. It also increases the pitch-line velocity, which makes the gears noisier and somewhat reduces the power transmitted, although not in direct proportion. In general, however, the use of more teeth to eliminate interference or undercutting is seldom an acceptable solution.

Another method of reducing interference and the resulting undercutting is to employ a larger pressure angle. The larger pressure angle creates smaller base circles, so that a greater portion of the tooth profile has an involute shape. In effect, this means that fewer teeth can be used; as a result, gears with a larger pressure angle are often smaller.

Of course, the use of standard gears is far less expensive than manufacturing specially made nonstandard gears. However, as indicated in Table 7.2, gears with larger pressure angles can be made without deviating from the standards.

One more way to eliminate interference is to use gears with shorter teeth. If the addendum distance is reduced, then points *C* and *D* move inward. One way to do this is to purchase standard gears and then grind the top lands of the teeth to a new addendum distance. This, of course, makes the gears nonstandard and causes concern about repair or replacement, but it can be effective in eliminating interference. Again, careful study of Table 7.2 indicates that this is also possible by use of standard 20° *stub tooth* gears.

7.8 CONTACT RATIO

The zone of action of meshing gear teeth is shown in Fig. 7.19, where tooth contact begins and ends at the intersections of the two addendum circles with the line of action. As always, initial contact occurs at *C*, and final contact occurs at *D*. Tooth profiles drawn through

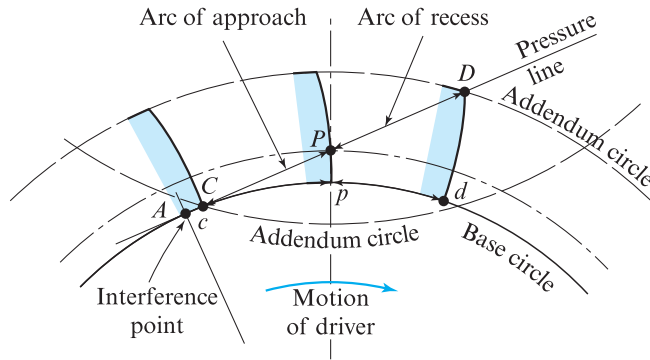


Figure 7.19

these points intersect the base circle at points c and d . Thinking back to our analogy of the unwrapping cord of Fig. 7.6, the linear distance CD , measured along the line of action, is equal to the arc length cd , measured along the base circle.

Consider a situation in which the arc length cd , or distance CD , is exactly equal to the base pitch, p_b , of Eq. (7.6). This means that one tooth and its tooth space spans the entire arc cd . In other words, when a tooth is just beginning contact at C , the tooth ahead of it is just ending its contact at D . Therefore, during the tooth action from C to D , there is exactly one pair of teeth in contact at all times.

Next, consider a situation for which the arc length cd , or distance CD , is greater than the base pitch, but not much greater, say $cd = 1.1p_b$. This means that when one pair of teeth is just entering contact at C , the previous pair, already in contact, has not yet reached D . Thus, for a short time, there are two pairs of teeth in contact, one in the vicinity of C , and the other nearing D . As meshing proceeds, the previous pair reaches D and ceases contact, leaving only a single pair of teeth in contact again, until the situation repeats itself with the next pair of teeth.

Because of the nature of this tooth action, with one, two, or even more pairs of teeth in contact simultaneously, it is convenient to define the term contact ratio as

$$m_c = \frac{CD}{p_b}. \quad (7.9)$$

This is a value for which the next lower integer indicates the average number of pairs of teeth in contact. Thus, a contact ratio of $m = 1.35$, for example, implies that there is always at least one tooth in contact, and there are two teeth in contact 35% of the time.

The minimum acceptable value of the contact ratio for smooth operation of meshing gears is $1.2 \leq m_c \leq 1.4$, and the recommended range of the contact ratio for most spur gearsets is $m_c > 1.4$.

The distance CD is quite convenient to measure if we are working graphically by making a drawing like Fig. 7.20 or Fig. 7.9. However, distances CP and PD can also be found analytically. From triangles O_3BC and O_3BP , we can write

$$CP = \sqrt{(R_3 + a)^2 - (R_3 \cos \phi)^2} - R_3 \sin \phi. \quad (7.10)$$

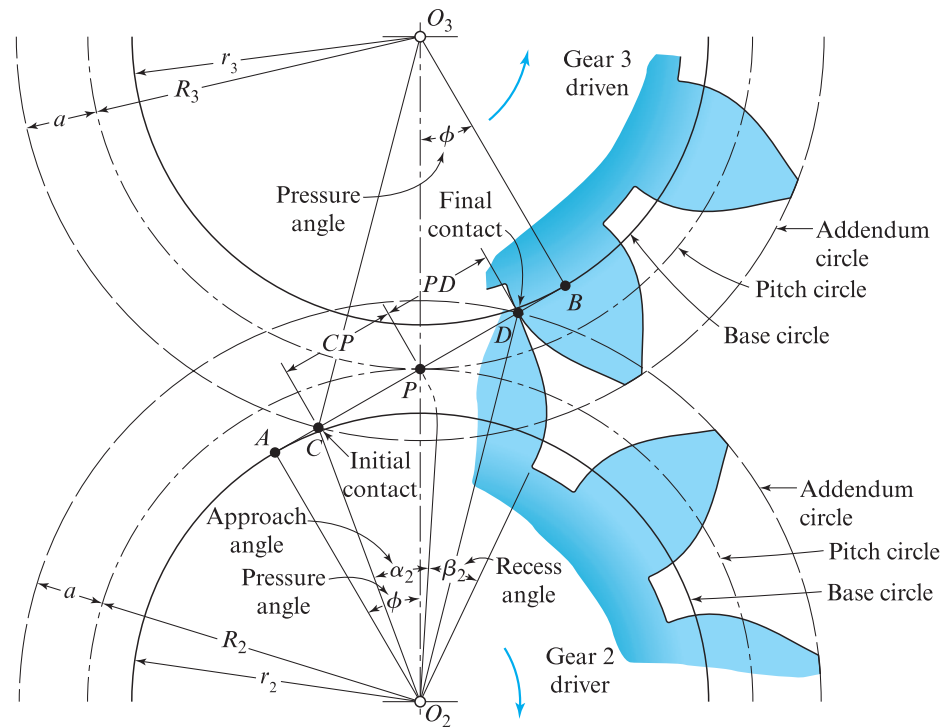


Figure 7.20

Similarly, from triangles O_2AD and O_2AP , we have

$$PD = \sqrt{(R_2 + a)^2 - (R_2 \cos \phi)^2} - R_2 \sin \phi. \quad (7.11)$$

The contact ratio is then obtained by substituting the sum of Eqs. (7.10) and (7.11) into Eq. (7.9).

We should note, however, that Eqs. (7.10) and (7.11) are only valid for the conditions where

$$CP \leq R_2 \sin \phi \quad \text{and} \quad PD \leq R_3 \sin \phi, \quad (7.12)$$

since proper contact cannot begin before point A or end after point B . If either of these inequalities is not satisfied, then the gear teeth have interference and undercutting results.

7.9 VARYING CENTER DISTANCE

Figure 7.21a shows a pair of meshing gears having 20° full-depth involute teeth. Since both sides of the teeth are in contact, the center distance $R_{O_3O_2}$ cannot be reduced without jamming or deforming the teeth. However, Fig. 7.21b shows the same pair of gears, but mounted with a slightly increased distance $R_{O_3O_2}$ between the shaft centers, as might happen through the accumulation of tolerances of surrounding parts. Clearance, or *backlash*, now exists between the teeth, as shown.

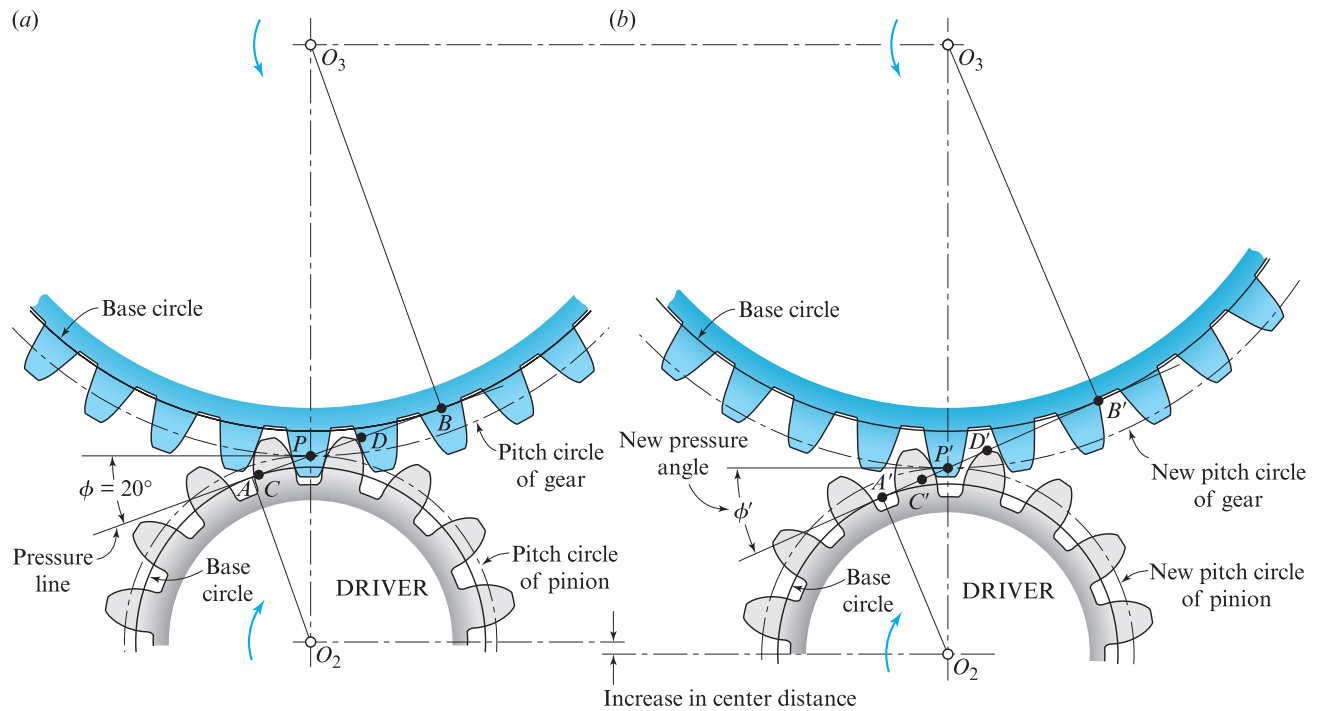


Figure 7.21 Effect of increased center distance on the action of involute gearing; mounting the gears at (a) normal center distance; (b) increased center distance.

When the center distance is increased, the base circles of the two gears do not change; they are fundamental to the shapes of the gears, once manufactured. However, review of Fig. 7.6 indicates that the same involute tooth shapes still touch as conjugate curves, and the fundamental law of gearing is still satisfied. However, the larger center distance results in an increase of the pressure angle and larger pitch circles passing through a new adjusted pitch point.

In Fig. 7.21b we see that triangles $O_2A'P'$ and $O_3B'P'$ are still similar to each other, although they are both modified by the change in pressure angle. Also, distances O_2A' and O_3B' are the base circle radii and have not changed. Therefore, the ratio of the new pitch radii, O_2P' and O_3P' , and the new velocity ratio remain the same as in the original design.

Another effect of increasing the center distance, observable in Fig. 7.21, is the shortening of the path of contact. The original path of contact CD in Fig. 7.21a is shortened to $C'D'$ in Fig. 7.21b. The contact ratio, Eq. (7.9), is also reduced when the path of contact $C'D'$ is shortened. Since a contact ratio of less than unity would imply periods during which no teeth would be in contact at all, the center distance must never be increased larger than that corresponding to a contact ratio of unity ($C'D' = p_b$).

7.10 INVOLUTOMETRY

Study of the geometry of the involute curve is called *involutometry*. In Fig. 7.22, a base circle with center at O is used to generate involute BC . AT is the generating line, ρ is the instantaneous radius of curvature of the involute, and r is the radius to point T on the

curve. If we designate the radius of the base circle R_b , line AT has the same length as the arc distance AB and so

$$\rho = r_b (\alpha + \varphi), \quad (a)$$

where α is the angle between the radius OT and the radius to the origin of the involute OB , and φ is the angle between the radius of the base circle OA and the radius OT . Since OAT is a right triangle,

$$\rho = r_b \tan \varphi. \quad (7.13)$$

Solving Eqs. (a) and (7.13) simultaneously to eliminate ρ and r_b gives

$$\alpha = \tan \varphi - \varphi,$$

which can be written

$$\text{inv } \varphi = \tan \varphi - \varphi \quad (7.14)$$

and defines the *involute function*. The angle φ in this equation is the variable involute angle, given in radians. Once φ is known, $\text{inv } \varphi$ can readily be determined from Eq. (7.14). The inverse problem, when $\text{inv } \varphi$ is given and φ is to be found, is more difficult. One approach is to expand Eq. (7.14) in an infinite series and to employ the first several terms to obtain a numeric approximation. Another approach is to use a root-finding technique [2, Chap. 4]. Here, we refer to Table 6 of Appendix A, in which the value of the involute function is tabulated, and the angle φ can be determined directly, in degrees.

Referring again to Fig. 7.22, we see that

$$r = \frac{r_b}{\cos \varphi}. \quad (7.15)$$

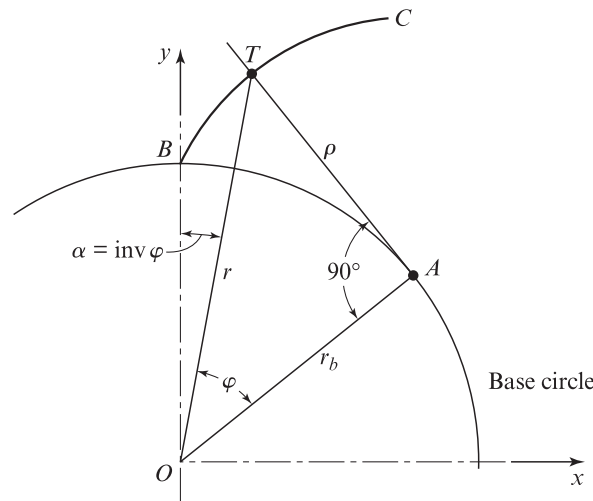


Figure 7.22

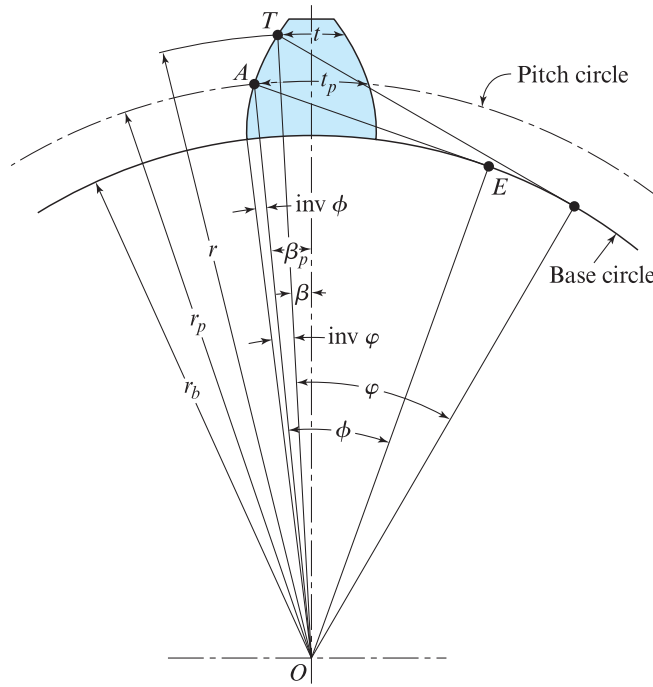


Figure 7.23

To illustrate the use of the relations obtained earlier, let us determine the tooth dimensions of Fig. 7.23. Here, only the portion of the tooth extending above the base circle has been drawn, and the thickness of the tooth, t_p , at the pitch circle (point A), equal to half of the circular pitch, is given. The problem is to determine the tooth thickness at some other point, say point T . The various quantities shown in Fig. 7.23 are identified as follows:

- r_b = radius of the base circle;
- r_p = radius of the pitch circle;
- r = radius at which the tooth thickness is to be determined;
- t_p = tooth thickness at the pitch circle;
- t = tooth thickness to be determined;
- ϕ = pressure angle corresponding to the pitch circle radius r_p ;
- φ = involute angle corresponding to point T ;
- β_p = angular half-tooth thickness at the pitch circle; and
- β = angular half-tooth thickness at point T .

The half-tooth thicknesses at points A and T are

$$\frac{t_p}{2} = \beta_p r_p \quad \text{and} \quad \frac{t}{2} = \beta r, \quad (b)$$

so that

$$\beta_p = \frac{t_p}{2r_p} \quad \text{and} \quad \beta = \frac{t}{2r}. \quad (c)$$

From these, we can write

$$\text{inv } \phi - \text{inv } \phi = \beta_p - \beta = \frac{t_p}{2r_p} - \frac{t}{2r}. \quad (d)$$

The tooth thickness at point T is obtained by solving Eq. (d) for t :

$$t = 2r \left(\frac{t_p}{2r_p} + \text{inv } \phi - \text{inv } \phi \right). \quad (7.16)$$

EXAMPLE 7.2

A gear has 22 teeth cut full depth with pressure angle $\phi = 20^\circ$ and diametral pitch $P = 2$ teeth/in. Find the thickness of the teeth at the base circle and at the addendum circle.

SOLUTION

By the equations of Sec. 7.1 and Table 7.2, we find the radius of the pitch circle is $R_p = R = N/2P = 5.500$ in, the circular pitch is $p = 1.571$ in/tooth, the addendum is $a = 0.500$ in, and the dedendum is $d = 0.625$ in.

From right triangle OEA in Fig. 7.23, the radius of the base circle can be written as

$$r_b = r_p \cos \phi = (5.500 \text{ in}) \cos 20^\circ = 5.168 \text{ in}.$$

The thickness of the tooth at the pitch circle is

$$t_p = \frac{p}{2} = \frac{1.571 \text{ in/tooth}}{2} = 0.7855 \text{ in}.$$

Converting the tooth pressure angle into radians gives $\phi = 20^\circ = 0.349066$ rad. Then, the involute function from Eq. (7.14) is

$$\text{inv } \phi = \tan 0.349066 - 0.349066 = 0.014904 \text{ rad}.$$

The involute angle at the base circle from Eq. (7.15), is $\phi_b = 0$. Therefore, the involute function is

$$\text{inv } \phi_b = 0.$$

Substituting these results into Eq. (7.16), the tooth thickness at the base circle is

$$t_b = 2r_b \left[\frac{t_p}{2r_p} + \text{inv } \phi - \text{inv } \phi_b \right] = 2(5.168 \text{ in}) \left[\frac{0.7855 \text{ in}}{2(5.500 \text{ in})} + 0.014904 - 0 \right] = 0.892 \text{ in}.$$

Ans.

The radius of the addendum circle is $r_a = r_p + a = 5.500 + 0.500 = 6.000$ in. Therefore, the involute pressure angle corresponding to this radius, from Eq. (7.15), is

$$\phi = \cos^{-1} \left(\frac{r_b}{r} \right) = \cos^{-1} \left(\frac{5.168 \text{ in}}{6.000 \text{ in}} \right) = 30.53^\circ = 0.532806 \text{ rad}.$$

Thus, the involute function is

$$\text{inv } \phi = \tan 0.532\,806 - 0.532\,806 = 0.056\,887 \text{ rad.}$$

Substituting these results into Eq. (7.16), the tooth thickness at the addendum circle is

$$\begin{aligned} t_a &= 2r_a \left[\frac{t_p}{2r_p} + \text{inv } \phi - \text{inv } \varphi \right] = 2(6.000 \text{ in}) \left[\frac{0.785\,5 \text{ in}}{2(5.500 \text{ in})} + 0.014\,904 - 0.056\,887 \right] \\ &= 0.353 \text{ in.} \end{aligned} \quad \text{Ans.}$$

Note that the tooth thickness at the base circle is more than double the tooth thickness at the addendum circle.

7.11 NONSTANDARD GEAR TEETH

In this section, we will examine the effects obtained by deviating from the specified standards and modifying such things as pressure angle, tooth depth, addendum, or center distance. Some of these modifications do not eliminate interchangeability; all of them are discussed with the intent of obtaining improved performance. Still, making such modifications probably means increased cost, since modified gears are available and must be specially machined for the particular application. Of course, it is also necessary at the time of any future repair or design modification.

The designer is often under great pressure to produce a design using gears that are small and yet able to transmit a large amount of power. Consider, for example, a gearset that must have a 4:1 velocity ratio. If the smallest pinion that will carry the load has a pitch diameter of 2 in, the mating gear has a pitch diameter of 8 in, making the overall space required for the two gears more than 10 in. On the other hand, if the pitch diameter of the pinion can be reduced by only $\frac{1}{4}$ in, the pitch diameter of the gear is reduced by a full 1 in, and the overall size of the gearset is reduced by $1\frac{1}{4}$ in. This reduction assumes considerable importance when it is realized that the associated machine elements, such as shafts, bearings, and enclosure, are also reduced in size.

If a tooth of a certain pitch is required to carry the load, the only method of decreasing the pinion diameter is to use fewer teeth. However, we have already seen that problems involving interference, undercutting, and contact ratio are encountered when the tooth numbers are made too small. Thus, three principal reasons for employing nonstandard gears are: (a) to eliminate undercutting, (b) to prevent interference, and (c) to maintain a reasonable contact ratio. Note also that if a pair of gears is manufactured of the same material, the pinion is the weaker and is subject to greater wear, since each of its teeth are in contact a greater portion of the time. Therefore, any undercutting weakens the tooth that is already weaker. Thus, another objective of nonstandard gears is to gain a better balance of strength between the pinion and the gear.

As an involute curve is generated from its base circle, its radius of curvature becomes larger and larger. Near the base circle, the radius of curvature is quite small, being theoretically zero at the base circle. Contact near this region of sharp curvature should be avoided if possible because of the difficulty in obtaining good cutting accuracy in areas

of small curvature, and since contact stresses are likely to be very high. Nonstandard gears present the opportunity of designing to avoid these sensitive areas.

Clearance Modification A larger fillet radius at the root of the tooth increases the fatigue strength of the tooth and provides extra depth for shaving the tooth profile. Since interchangeability is not lost, the dedendum is sometimes increased to $1.300/P$ or $1.400/P$ to obtain space for a larger fillet radius.

Center-Distance Modification When gears of low tooth numbers are paired with each other or with larger gears, reduction in interference and improvement in the contact ratio can be obtained by increasing the center distance to greater than standard. Although such a system changes the tooth proportions and the pressure angle of the gears, the resulting tooth shapes can be generated with rack cutters (or hobs) of standard pressure angles or with standard pinion shapers. Before introducing this system, however, it will be of value to develop certain additional relations about the geometry of gears.

The first new relation is for finding the thickness of a tooth that is cut by a rack cutter (or hob) when the pitch line of the rack cutter is displaced or offset a distance, e , from the pitch circle of the gear being cut. What we are doing here is moving the rack cutter away from the center of the gear being cut. Stated another way, suppose the rack cutter does not cut as deeply into the gear blank, and the teeth are not cut to full depth. This produces teeth that are thicker than the standard, and this thickness will now be determined. Figure 7.24*a* shows the problem, and Fig. 7.24*b* shows the solution. The increase of tooth thickness at the pitch circle is $2e \tan \phi$, so that

$$t = 2e \tan \phi + \frac{p}{2}, \quad (7.17)$$

where ϕ is the pressure angle of the rack cutter and t is the thickness of the modified gear tooth measured on its modified pitch circle.

Now suppose that two gears of different tooth numbers have both been cut with the cutter offset from their pitch circles as in the previous paragraph. Since the teeth of both have been cut with offset cutters, they will mate at a modified pressure angle and with modified pitch circles and consequently modified center distances. The word *modified* is used here in the sense of being nonstandard. Our problem is to determine the radii of these modified pitch circles and the value of the modified pressure angle.

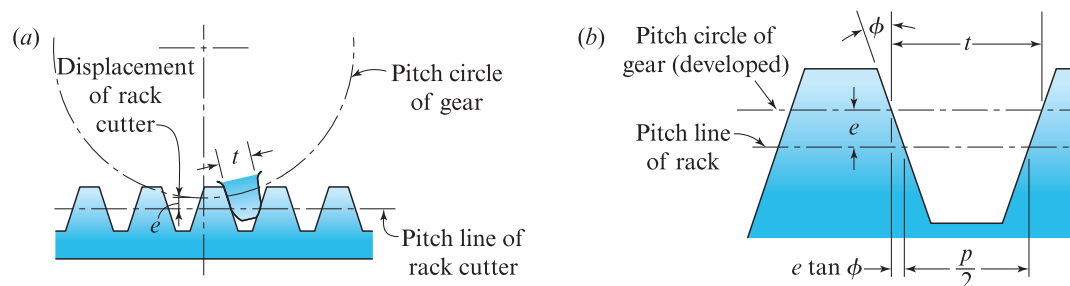


Figure 7.24

In the following notation, the word *standard* refers to values that would have been obtained had the usual, or standard, systems been employed to obtain the dimensions:

- ϕ = standard pressure angle of generating rack cutter;
- ϕ' = modified pressure angle at which gears will mate;
- R_2 = standard pitch radius of pinion;
- R'_2 = modified pitch radius of pinion when meshing with given gear;
- R_3 = standard pitch radius of gear;
- R'_3 = modified pitch radius of gear when meshing with given pinion;
- t_2 = thickness of pinion tooth at standard pitch radius R_2 ;
- t'_2 = thickness of pinion tooth at modified pitch radius R'_2 ;
- t_3 = thickness of gear tooth at standard pitch radius R_3 ; and
- t'_3 = thickness of gear tooth at modified pitch radius R'_3 .

From Eq. (7.16), the thickness of a gear tooth at the standard pitch radius and at the modified pitch radius can be written, respectively, as

$$t'_2 = 2R'_2 \left(\frac{t_2}{2R_2} + \text{inv } \phi - \text{inv } \phi' \right) \quad (a)$$

and

$$t'_3 = 2R'_3 \left(\frac{t_3}{2R_3} + \text{inv } \phi - \text{inv } \phi' \right). \quad (b)$$

Note that the sum of these two thicknesses must be the new circular pitch. Therefore, using Eq. (7.3), we can write

$$t'_2 + t'_3 = p' = \frac{2\pi R'_2}{N_2}. \quad (c)$$

Since the pitch diameters of a pair of mating gears are proportional to their tooth numbers, then

$$R_3 = \frac{N_3}{N_2} R_2 \quad \text{and} \quad R'_3 = \frac{N_3}{N_2} R'_2. \quad (d)$$

Substituting Eqs. (a), (b), and (d) into Eq. (c) and rearranging gives

$$\text{inv } \phi' = \frac{N_2 (t'_2 + t'_3) - 2\pi R_2}{2R_2 (N_2 + N_3)} + \text{inv } \phi. \quad (7.18)$$

This equation gives the modified pressure angle ϕ' , at which a pair of gears will operate when the tooth thicknesses on their standard pitch circles are modified to t'_2 and t'_3 .

Although the base circle of a gear is fundamental to its shape and fixed once the gear is generated, gears have no pitch circles until a pair of them are brought into contact. Bringing a pair of gears into contact creates a pair of pitch circles that are tangent to each other at the modified pitch point. Throughout this discussion, the idea of a pair of so-called standard pitch circles has been used to define a certain point on the involute curves. These standard pitch circles, as we have seen, are the ones that would come into existence when

the gears are paired *if the gears are not modified from the standard dimensions*. On the other hand, the base circles are fixed circles that are not changed by tooth modifications. The base circle remains the same whether the tooth dimensions are changed or not, so we can determine the base circle radius using either the standard pitch circle or the new pitch circle. Thus, from Eq. (7.15), we can write

$$R_2 \cos \phi = R'_2 \cos \phi'.$$

Therefore, the modified pitch radius of the pinion can be written as

$$R'_2 = \frac{R_2 \cos \phi}{\cos \phi'}. \quad (7.19)$$

Similarly, the modified pitch radius of the gear can be written as

$$R'_3 = \frac{R_3 \cos \phi}{\cos \phi'}. \quad (7.20)$$

Equations (7.19) and (7.20) give the values of the actual pitch radii when the two gears with modified teeth are brought into mesh without backlash. The new center distance is, of course, the sum of these radii.

All the necessary relations have now been developed to create nonstandard gears with changes in the center distance. The use of these relations is now illustrated by an example.

Figure 7.25 is a drawing of a 20° pressure angle, 1-tooth/in diametral pitch, 12-tooth pinion generated with a rack cutter to full depth with a standard clearance of $0.250/P$. In the 20° full-depth system, interference is severe when the number of teeth is less than 14. The resulting undercutting is evident in the drawing.

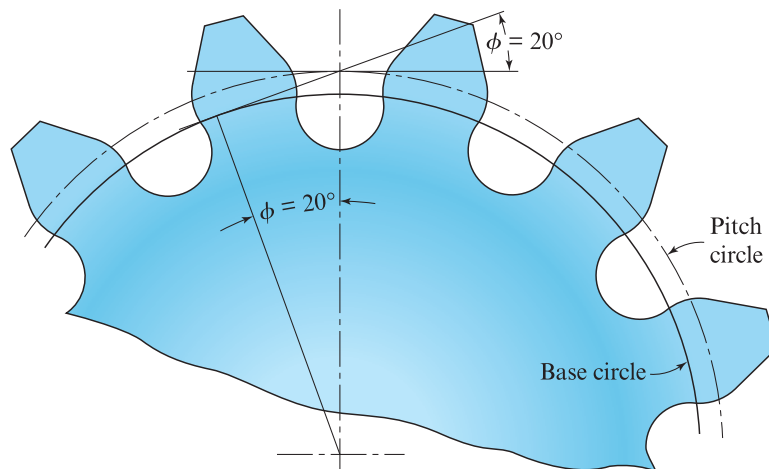


Figure 7.25 Standard 20° pressure angle, 1-tooth/in diametral pitch, 12-tooth full-depth involute gear showing undercut.

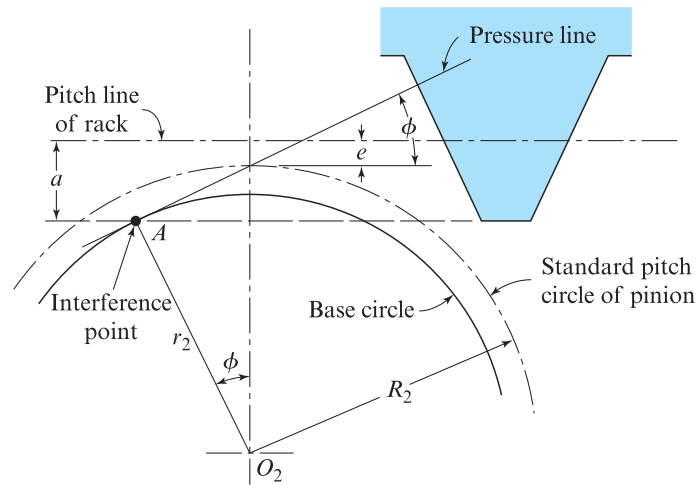


Figure 7.26 Offset of a rack cutter to cause its addendum line to pass through the interference point.

In an attempt to eliminate undercutting, improve the tooth action, and increase the contact ratio, suppose that this pinion were not cut to full depth; suppose instead that the rack cutter were only allowed to cut to a depth for which its addendum passes through the interference point A of the pinion being cut, that is—the point of tangency of the 20° line of action and the base circle—as shown in Fig. 7.26. From Eq. (7.15) we know that

$$r_2 = R_2 \cos \phi. \quad (e)$$

Then, from Fig. 7.26, the depth of the cut would be offset from the standard by

$$e = a + r_2 \cos \phi - R_2. \quad (f)$$

Substituting Eq. (e) into Eq. (f), the offset can be written as

$$e = a + R_2 \cos^2 \phi - R_2 = a - R_2 \sin^2 \phi. \quad (7.21)$$

If the offset is any less than this, then the rack will cut below the interference point, A , and will result in undercutting.

EXAMPLE 7.3

A 12-tooth pinion with pressure angle $\phi = 20^\circ$ and diametral pitch $P = 1$ tooth/in is to be mated with a standard 40-tooth gear. If the pinion were cut to full depth, then Eq. (7.9) demonstrates that the contact ratio would be 1.41, but there would be undercutting, as indicated in Fig. 7.25. Instead, let the 12-tooth pinion be cut from a larger blank using center-distance modifications. Determine the cutter offset, the modified pressure angle, the modified pitch radii of the pinion and the gear, the modified center distance, the modified outside radii of the pinion and gear, and the contact ratio. Has the contact ratio increased significantly?

SOLUTION

Designating the pinion as subscript 2 and the gear as 3, then with $P = 1$ tooth/in and $\phi = 20^\circ$, the following values are determined:

$$p = 3.142 \text{ in/tooth}, R_2 = 6 \text{ in}, R_3 = 20 \text{ in}, N_2 = 12 \text{ teeth}, \\ N_3 = 40 \text{ teeth}, \quad \text{and} \quad t_3 = 1.571 \text{ in}.$$

For a standard rack cutter, from Table 7.2, the addendum is $a = 1/P = 1.0$ in.

From Eq. (7.21), the rack cutter will be offset by

$$e = 1.0 - 6.0 \sin^2 20^\circ = 0.298 \text{ in.} \quad \text{Ans.}$$

Then, the thickness of the pinion tooth at the 6-in pitch circle, using Eq. (7.17), is

$$t'_2 = 2e \tan \phi + \frac{p}{2} = 2(0.298 \text{ in}) \tan 20^\circ + \frac{3.142 \text{ in}}{2} = 1.788 \text{ in}.$$

The pressure angle at which this (and only this) gearset will operate is determined from Eq. (7.18); that is,

$$\begin{aligned} \text{inv } \phi' &= \frac{N_2 (t'_2 + t'_3) - 2\pi R_2}{2R_2 (N_2 + N_3)} + \text{inv } \phi \\ &= \frac{12(1.788 \text{ in} + 1.571 \text{ in}) - 2\pi(6.0 \text{ in})}{2(6.0 \text{ in})(12 + 40)} + \text{inv } 20^\circ = 0.01908 \text{ rad.} \end{aligned}$$

From Appendix A, Table 6, we find that the new pressure angle is

$$\phi' = 21.65^\circ. \quad \text{Ans.}$$

Using Eqs. (7.19) and (7.20), the modified pitch radii are found to be

$$R'_2 = \frac{R_2 \cos \phi}{\cos \phi'} = \frac{(6.0 \text{ in}) \cos 20^\circ}{\cos 21.65^\circ} = 6.066 \text{ in}, \quad \text{Ans.}$$

$$R'_3 = \frac{R_3 \cos \phi}{\cos \phi'} = \frac{(20.0 \text{ in}) \cos 20^\circ}{\cos 21.65^\circ} = 20.220 \text{ in.} \quad \text{Ans.}$$

So, the modified center distance is

$$R'_2 + R'_3 = 6.066 + 20.220 = 26.286 \text{ in.} \quad \text{Ans.}$$

Note that the center distance has not increased as much as the offset of the rack cutter.

Standard clearance of $0.25/P$ results from the standard dedendum equal to $1.25/P$, as indicated in Table 7.2. So, the root radii of the two gears are

$$\text{Root radius of pinion} = 6.298 - 1.250 = 5.048 \text{ in},$$

$$\text{Root radius of gear} = 20.000 - 1.250 = 18.750 \text{ in},$$

$$\text{Sum of root radii} = 23.798 \text{ in}.$$

The difference between this sum and the center distance is the working depth plus twice the clearance. Since the clearance is 0.25 in for each gear, the working depth is

$$\text{Working depth} = 26.286 - 23.798 - 2(0.250) = 1.988 \text{ in.}$$

The outside radius of each gear is the sum of the root radius, the clearance, and the working depth; that is,

$$\text{Outside radius of pinion} = 5.048 + 0.250 + 1.988 = 7.286 \text{ in.;} \quad \text{Ans.}$$

$$\text{Outside radius of gear} = 18.750 + 0.250 + 1.988 = 20.988 \text{ in.} \quad \text{Ans.}$$

The result is shown in Fig. 7.27, and the pinion is seen to have a stronger-looking form than the one of Fig. 7.25. Undercutting has been completely eliminated.

The contact ratio can be obtained from Eqs. (7.9) through (7.11). The following quantities are needed:

$$\text{Outside radius of pinion} = R'_2 + a = 7.286 \text{ in,}$$

$$\text{Outside radius of gear} = R'_3 + a = 20.988 \text{ in,}$$

$$r_2 = R_2 \cos \phi = (6.000 \text{ in}) \cos 20^\circ = 5.638 \text{ in,}$$

$$r_3 = R_3 \cos \phi = (20.000 \text{ in}) \cos 20^\circ = 18.794 \text{ in,}$$

$$p_b = p \cos \phi = (3.1416 \text{ in/tooth}) \cos 20^\circ = 2.952 \text{ in/tooth.}$$

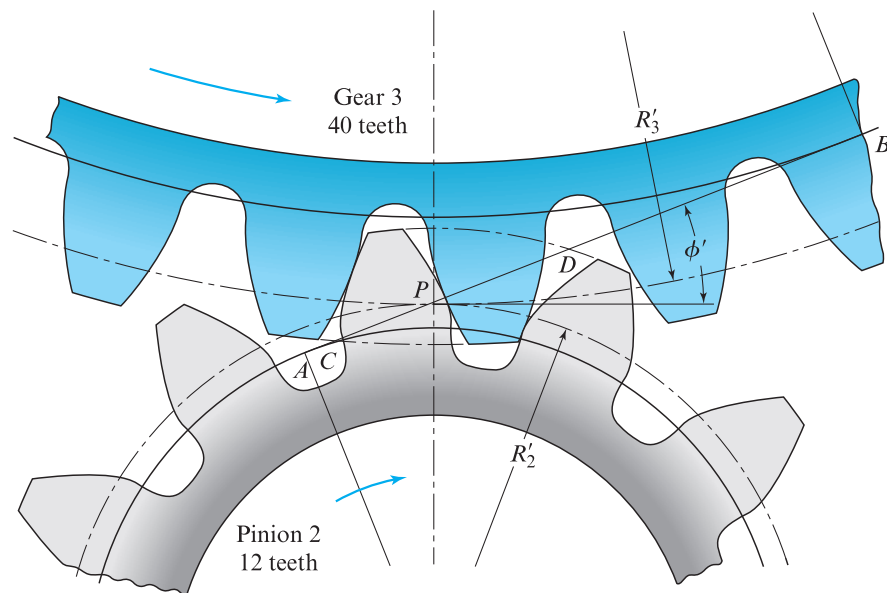


Figure 7.27

Therefore, for Eqs. (7.10) and (7.11), we have

$$\begin{aligned}
 CP &= \sqrt{(R'_3 + a)^2 - r_3^2} - R'_3 \sin \phi \\
 &= \sqrt{(20.988 \text{ in})^2 - (18.794 \text{ in})^2} - (20.220 \text{ in}) \sin 21.65^\circ = 1.883 \text{ in}, \\
 PD &= \sqrt{(R'_2 + a)^2 - r_2^2} - R'_2 \sin \phi \\
 &= \sqrt{(7.286 \text{ in})^2 - (5.638 \text{ in})^2} - (6.066 \text{ in}) \sin 21.65^\circ \\
 &= 2.377 \text{ in}.
 \end{aligned}$$

Finally, from Eq. (7.9), the contact ratio is

$$m_c = \frac{CP + PD}{p_b} = \frac{1.883 \text{ in} + 2.377 \text{ in}}{2.952 \text{ in/tooth}} = 1.443 \text{ teeth avg.} \quad \text{Ans.}$$

Therefore, the contact ratio has increased only slightly (approximately 2% increase). The modification, however, is justified because of the elimination of undercutting, which results in a substantial improvement in the strength of the teeth.

Long-and-Short-Addendum System It often happens in the design of machinery that the center distance between a pair of gears is fixed by some other design consideration or feature of the machine. In such a case, modifications to obtain improved performance cannot be made by varying the center distance.

In the previous section, we saw that improved action and tooth shape can be obtained by backing the rack cutter away from the gear blank during forming of the teeth. The effect of this withdrawal is to create the active tooth profile farther away from the base circle. Examination of Fig. 7.27 indicates that more dedendum could be used on the gear (not the pinion) before the interference point is reached. If the rack cutter is advanced into the gear blank by a distance equal to the withdrawal from the pinion blank, more of the gear dedendum is used, and at the same time, the center distance is not changed. This is called the *long-and-short-addendum system*.

In the long-and-short-addendum system, there are no changes in the pitch circles and consequently none in the pressure angle. The effect is to move the contact region away from the pinion center toward the gear center, thus shortening the approach action and lengthening the recess action.

The characteristics of the long-and-short-addendum system can be explained by reference to Fig. 7.28. Figure 7.28*a* shows a conventional (standard) set of gears having a dedendum equal to the addendum plus the clearance. Interference exists, and the tip of the gear tooth will have to be relieved as shown or the pinion will be undercut. This is indicated, since the addendum circle crosses the line of action at *C*, outside of the tangency or interference point *A*; hence, distance *AC* is a measure of the degree of interference.

To eliminate the undercutting or interference, the pinion addendum may be enlarged, as in Fig. 7.28*b*, until the addendum circle of the pinion passes through the interference point (point *B*) of the gear. In this manner, we shall be using all of the gear-tooth profile.

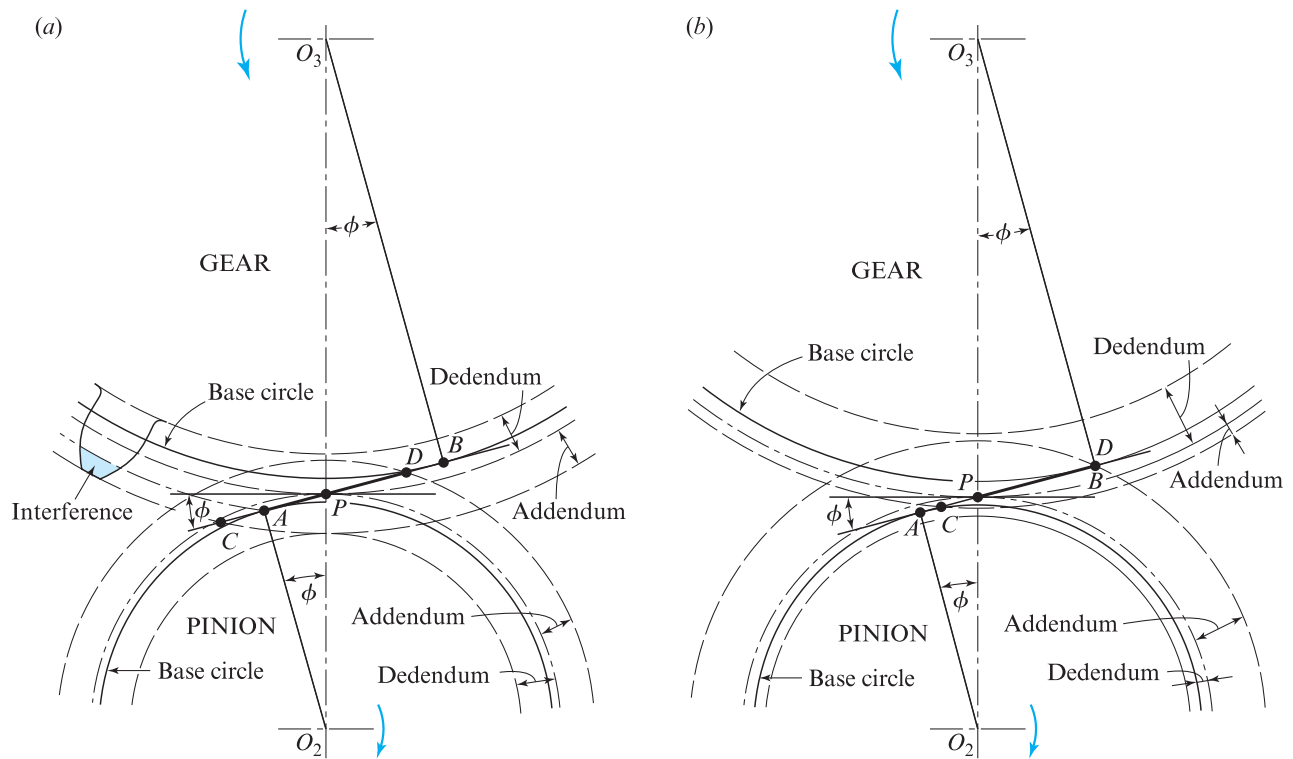


Figure 7.28 Comparison of standard gears and gears cut by the long-and-short-addendum system: (a) gear and pinion with standard addendum and dedendum; (b) gear and pinion with long-and-short addendum.

The same whole depth may be retained; hence, the dedendum of the pinion may be reduced by the same amount that the addendum is increased. This means that we must also lengthen the gear dedendum and shorten the dedendum of the mating pinion. With these changes, the path of contact is line CD of Fig. 7.28b. It is longer than path AD of Fig. 7.28a, and so the contact ratio is higher. Note, too, that the base circles, the pitch circles, the pressure angle, and the center distance have not changed. Both gears can be cut with standard cutters by advancing the cutter into the gear blank for this modification by a distance equal to the amount of withdrawal from the pinion blank. Finally, note that the blanks from which the pinion and gear are cut must now be of different diameters than the standard blanks.

The tooth dimensions for the long-and-short-addendum system can be determined using the equations developed in the previous sections.

A less obvious advantage of the long-and-short-addendum system is that more recess action than approach action is obtained. The approach action of gear teeth is analogous to pushing a piece of chalk across a blackboard; the chalk screeches. But, when the chalk is pulled across a blackboard, analogous to the recess action, it glides smoothly. Thus, recess action is always preferable because of the smoothness and the lower frictional forces.

7.12 PARALLEL-AXIS GEAR TRAINS

Mechanisms arranged in combinations so that the driven member of one mechanism is the driver for another mechanism are called *mechanism trains*. With certain exceptions, to be

explored here, the analysis of such trains can proceed in serial fashion by using the methods developed in the previous chapters.

In Chap. 3, we learned that the first-order *kinematic coefficient* is the term used to describe the ratio of the angular velocity of the driven member to that of the driving member. Thus, for example, in a four-bar linkage with link 2 as the driving or input member and link 4 as the driven or output member, we have

$$\theta'_{42} = \frac{d\theta_4}{d\theta_2} = \frac{d\theta_4/dt}{d\theta_2/dt} = \frac{\omega_4}{\omega_2}, \quad (a)$$

where it is noted that, as in Sec. 5.5 and after, we adopt the second subscript to explicitly indicate the number of the driving or input member. This second subscript is important in this and following sections, since many mechanism trains have more than one degree of freedom.

In this section, where we deal with serially connected gear trains, we prefer to write Eq. (a) as

$$\theta'_{LF} = \frac{d\theta_L}{d\theta_F} = \frac{d\theta_L/dt}{d\theta_F/dt} = \frac{\omega_L}{\omega_F}, \quad (7.22)$$

where ω_L is the angular velocity of the *last* gear and ω_F is the angular velocity of the *first* gear in the train, since, usually, the last gear is the output and is the driven gear, and the first is the input and driving gear.

The term θ'_{LF} in Eq. (7.22) is the first-order kinematic coefficient, called the *speed ratio* by some or the *train value* by others. Equation (7.22) is often written in the more convenient form:

$$\omega_L = \theta'_{LF} \omega_F. \quad (7.23)$$

Next, we consider pinion 2 driving gear 3. The speed of the driven gear is

$$\omega_3 = \pm \frac{R_2}{R_3} \omega_2 = \pm \frac{N_2}{N_3} \omega_2, \quad (b)$$

where, for each gear, R is the radius of the pitch circle, N is the number of teeth, and ω is either the angular velocity or the angular displacement completed during a chosen time interval.

For parallel-shaft gearing, the directions can be tracked by following the vector sense—that is, by specifying that angular velocity is positive when counterclockwise as seen from a chosen side. For parallel-shaft gearing, we shall use the following sign convention: If the last gear of a parallel-shaft gear train rotates with the same sense as the first gear, then θ'_{LF} is positive; if the last gear rotates in the opposite sense to the first gear, then θ'_{LF} is negative. This sign convention approach is not as easy, however, when the gear shafts are not parallel, as in bevel, crossed-helical, or worm gearing (Chap. 8). In such cases, it is often simpler to track the directions by visually inspecting a sketch of the train.

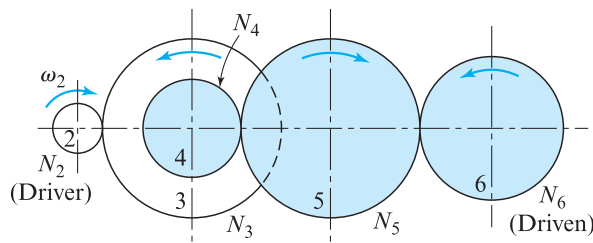


Figure 7.29

The gear train shown in Fig. 7.29 is made up of five gears in series. Applying Eq. (b) three times, we find the speed of gear 6 to be

$$\omega_6 = -\frac{R_5 R_4 R_2}{R_6 R_5 R_3} \omega_2 = -\frac{N_5 N_4 N_2}{N_6 N_5 N_3} \omega_2. \quad (c)$$

Here, we note that gear 5 is an *idler*; that is, its tooth numbers cancel in Eq. (c), and hence the only purpose served by gear 5 is to change the direction of rotation of gear 6. We further note that gears 5, 4, and 2 are drivers, whereas gears 6, 5, and 3 are driven members. Thus, Eq. (7.22) can also be written

$$\theta'_{LF} = \pm \frac{\text{product of driving tooth numbers}}{\text{product of driven tooth numbers}}. \quad (7.24)$$

Note also that, since they are proportional, pitch radii can be used in Eq. (7.24) just as well as tooth numbers.

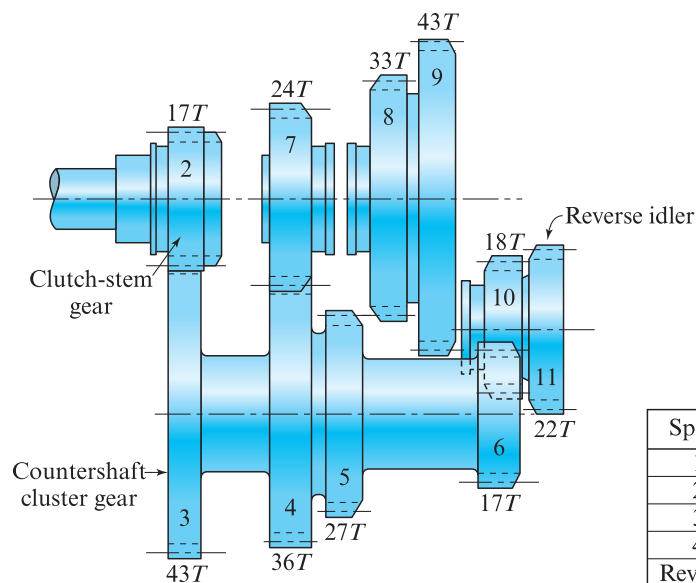


Figure 7.30 A truck transmission with gears having diametral pitch of 7 teeth/in and pressure angle of 22.5° .

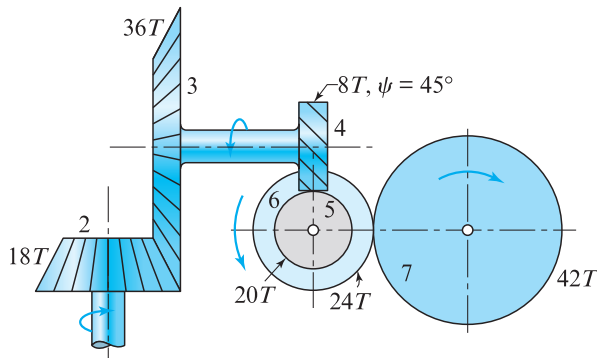


Figure 7.31 A gear train composed of bevel, crossed-helical, and spur gears.

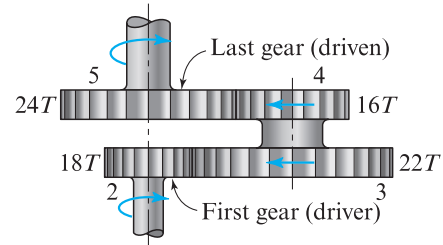


Figure 7.32 A reverted gear train.

In speaking of gear trains, it is convenient to describe a train having only one gear on each axis as a *simple* gear train. A *compound* gear train, then, is one that has two or more gears on one or more axes, such as the train shown in Fig. 7.29. Another example of a compound gear train is shown in Fig. 7.30. Figure 7.30 shows a transmission for a small- or medium-size truck that has four speeds forward and one in reverse.

The compound gear train shown in Fig. 7.31 is composed of bevel, helical (Chap. 8), and spur gears. The helical gears are crossed, so their direction of rotation depends upon their hand.

A *reverted* gear train is one in which the first and last gears have collinear axes of rotation, such as the one shown in Fig. 7.32. This produces a compact arrangement and is used in such applications as speed reducers, clocks (to connect the hour hand to the minute hand), and machine tools. As an exercise, it is suggested that you seek out a suitable set of diametral pitches for each pair of gears shown in Fig. 7.32 so that the first and last gears have the same axis of rotation with all gears properly engaged.

7.13 DETERMINING TOOTH NUMBERS

When notable power is transmitted through a speed reduction unit, the speed ratio of the last pair of meshing gears is usually chosen larger than that of the first gear pair, since the torque is greater at the low-speed end. In a given amount of space, more teeth can be used on gears of lesser pitch; hence, a greater speed reduction can be obtained at the high-speed end.

Without examining the problem of tooth strength, suppose we wish to use two pairs of gears in a train to obtain an overall kinematic coefficient of $\theta'_{LF} = \frac{1}{12}$. Let us also impose the restriction that the tooth numbers must not be less than 15 and that the reduction in the first pair of gears should be about twice that of the second pair. This means that the overall kinematic coefficient is

$$\theta'_{52} = \frac{N_4}{N_5} \frac{N_2}{N_3} = \frac{1}{12}, \quad (a)$$

where N_2/N_3 is the coefficient of the first gear pair, and N_4/N_5 is that of the second pair. Since the kinematic coefficient of the first pair should be half that of the second, Eq. (a) can be written as

$$\left(\frac{N_4}{N_5}\right) \left(\frac{N_2}{2N_3}\right) = \frac{1}{12} \quad (b)$$

or

$$\frac{N_4}{N_5} = \sqrt{\frac{1}{6}} = 0.408\ 248 \quad (c)$$

to six decimal places. The following tooth numbers are seen to be close:

$$\frac{15}{37} \quad \frac{16}{39} \quad \frac{18}{44} \quad \frac{20}{49} \quad \frac{22}{54} \quad \frac{24}{59}.$$

Of these, $N_4/N_5 = 20/49$ is the closest approximation, but note that

$$\theta'_{52} = \left(\frac{N_4}{N_5}\right) \left(\frac{N_2}{N_3}\right) = \left(\frac{20}{49}\right) \left(\frac{20}{98}\right) = \frac{400}{4\ 802} = \frac{1}{12.005},$$

which is very close to $1/12$. On the other hand, the choice of $N_4/N_5 = 18/44$ gives exactly

$$\theta'_{52} = \left(\frac{N_4}{N_5}\right) \left(\frac{N_2}{N_3}\right) = \left(\frac{18}{44}\right) \left(\frac{22}{108}\right) = \frac{396}{4\ 752} = \frac{1}{12}.$$

In this case, the reduction in the first gear pair is not exactly twice the reduction in the second gear pair. However, this consideration is usually of only minor importance.

The problem of specifying tooth numbers and the number of pairs of gears to give a kinematic coefficient with a specified degree of accuracy has interested many people, but has no exact solution. Consider, for instance, the problem of specifying a set of gears to have a kinematic coefficient of $\theta'_{LF} = \pi/10$ accurate to eight decimal places, while we know that π is an irrational number and cannot be expressed as a ratio of integers.

7.14 EPICYCLIC GEAR TRAINS

Figure 7.33 shows an elementary epicyclic gear train together with its schematic diagram, as suggested by Lévai.* The train consists of a *central gear*, 2, and an *epicyclic gear*, 4, which produces epicyclic motion for its points by rolling around the periphery of the central gear. An *arm* 3 contains the bearing for the epicyclic gear to maintain the gears in mesh.

* Literature devoted to epicyclic gear trains is rather scarce; however, see [3]. For a comprehensive study in the English language, see [1]; this book lists 104 references.

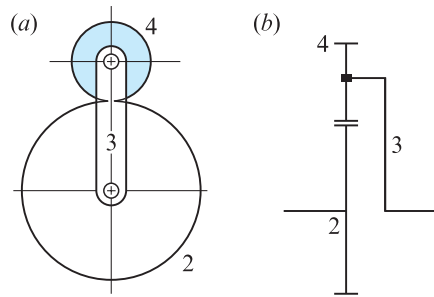


Figure 7.33 (a) Elementary epicyclic gear train; (b) its schematic diagram.

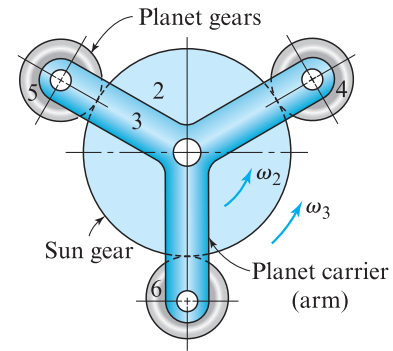


Figure 7.34 A planetary gearset.

Epicyclic trains are also called *planetary* or *sun-and-planet* gear trains. In this nomenclature, gear 2 of Fig. 7.33 is called the *sun gear*, gear 4 is called the *planet gear*, and crank 3 is called the *planet carrier*. Figure 7.34 shows the train of Fig. 7.33 with two redundant planet gears added. This produces better force balance; also, adding more planet gears allows lower forces by more load sharing. However, these additional planet gears do not change the kinematic characteristics at all. For this reason, we generally indicate only a single planet in the illustrations and problems in Chap. 7, although an actual machine will probably be designed with planets in trios.

The simple epicyclic gear train together with its schematic designation in Fig. 7.35 shows how the motion of the planet gear can be transmitted to another central gear. The second central gear in this case is gear 5, an internal gear. In Fig. 7.35a, internal gear 5 is stationary, but this is not a requirement, as shown in Fig. 7.35b. Figure 7.36 shows a similar arrangement, with the difference that both central gears are external gears. Note, in Fig. 7.36, that the double planet gears are mounted on a single planet shaft and that each planet gear is in mesh with a separate sun gear rotating at a different speed.

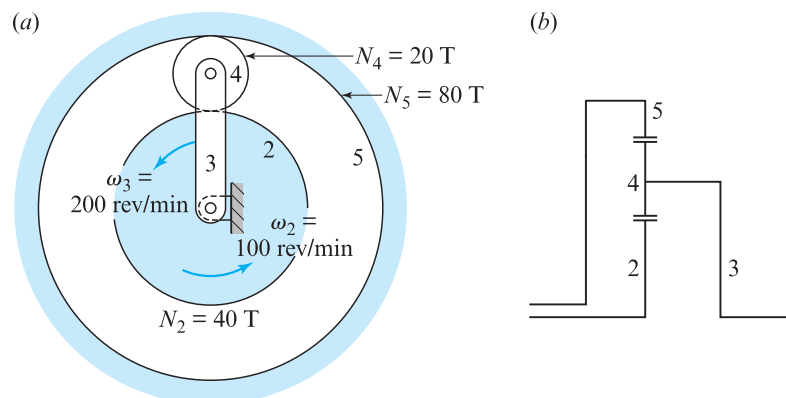


Figure 7.35 (a) A simple epicyclic gear train; (b) its schematic diagram.

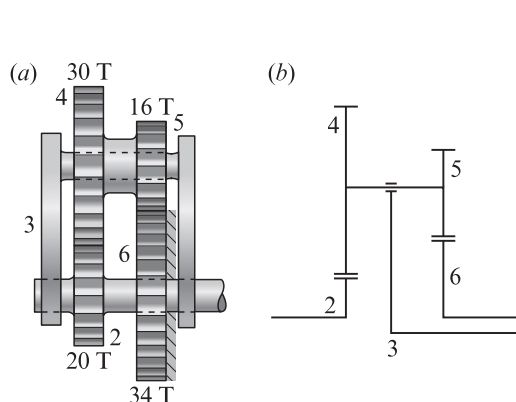


Figure 7.36 A simple epicyclic gear train with double planet gears.

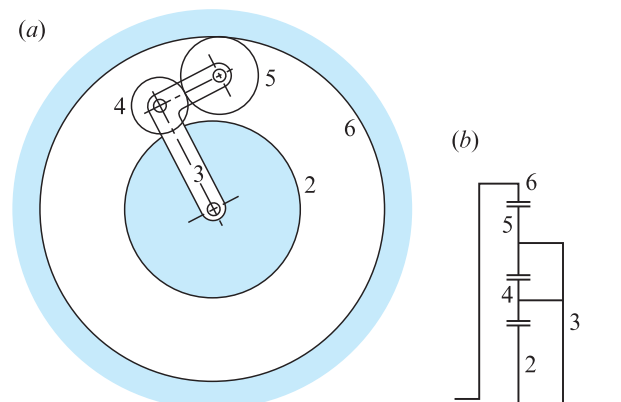


Figure 7.37 An epicyclic gear train with two planet gears.

In any case, no matter how many planets are used, only one planet carrier or arm may be used. This principle is shown in Fig. 7.34, in which redundant planets are used, and in Fig. 7.37, where two planets are used to alter the kinematic performance.

According to Lévai, only twelve variations of epicyclic gear trains are possible; they are all shown in schematic form in Fig. 7.38 as Lévai arranged them. In all variations, the arm (the planet carrier) is shown as link number 3. The trains in Figs. 7.38a and 7.38c are simple trains in which the planet gears mesh with both sun gears. The trains shown in Figs. 7.38b and 7.38d have planet gear pairs that are partly in mesh with each other and partly in mesh with the sun gears.

7.15 ANALYSIS OF EPICYCLIC GEAR TRAINS BY FORMULA

Figure 7.39 shows a planetary gear train composed of a sun gear 2, an arm or planet carrier, 3, and planet gears 4 and 5. Using the apparent angular velocity equation, Eq. (3.10), we can write that the angular velocity of gear 2 as it would appear from a coordinate system fixed to arm 3 is

$$\omega_{2/3} = \omega_2 - \omega_3. \quad (a)$$

Also, the angular velocity of gear 5 as it would appear from the coordinate system fixed to arm 3 is

$$\omega_{5/3} = \omega_5 - \omega_3. \quad (b)$$

Dividing Eq. (b) by Eq. (a) gives

$$\frac{\omega_{5/3}}{\omega_{2/3}} = \frac{\omega_5 - \omega_3}{\omega_2 - \omega_3}. \quad (c)$$

Equation (c) expresses the ratio of the apparent angular velocity of gear 5 to that of gear 2, with both taken as they would appear from arm 3. This ratio, which is proportional to

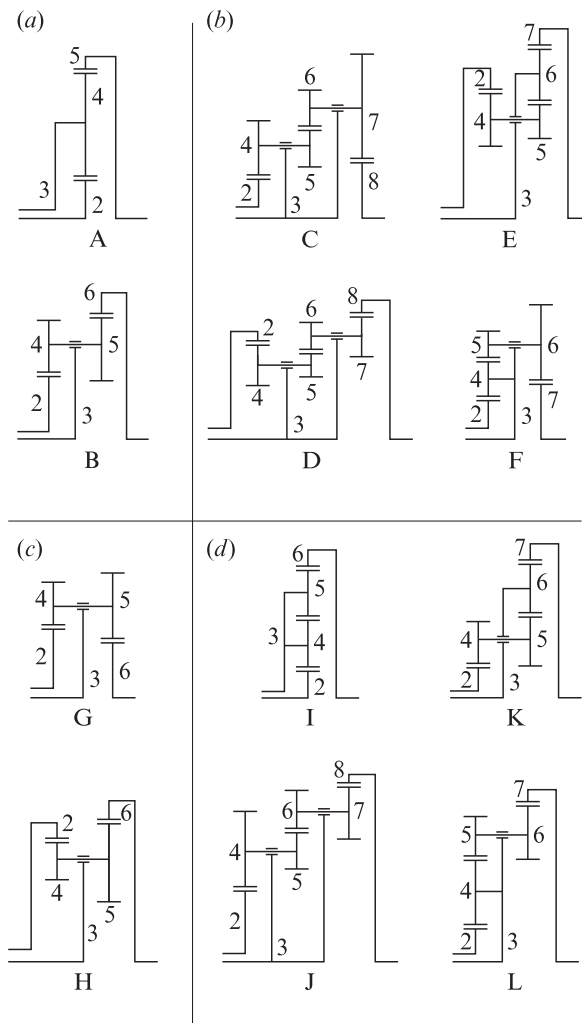
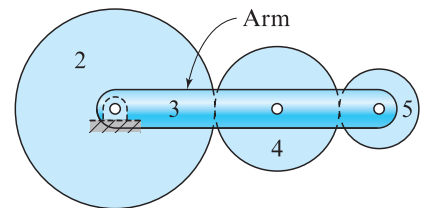


Figure 7.38 All twelve possible epicyclic gear train types according to Lévai [1]. (Reproduced with permission of the author.)

Figure 7.39



the tooth numbers, appears the same whether the arm is rotating or not; it is the first-order kinematic coefficient of the gear train. Therefore, the first-order kinematic coefficient can be expressed as

$$\theta'_{52/3} = \frac{\omega_5 - \omega_3}{\omega_2 - \omega_3}. \quad (d)$$

An equation similar to Eq. (d) is all that we need to find the angular velocities in a planetary gear train. It is convenient to express it in the form as it would appear from the arm:

$$\theta'_{LF/A} = \frac{\omega_L - \omega_A}{\omega_F - \omega_A}, \quad (7.25)$$

where

ω_F = angular velocity of the first gear in the train,
 ω_L = angular velocity of the last gear in the train, and
 ω_A = angular velocity of the arm.

Note that Eq. (d) and Eq. (7.25) can be expressed entirely in terms of kinematic coefficients; that is,

$$\theta'_{52/3} = \frac{\theta'_5 - \theta'_3}{\theta'_2 - \theta'_3} \quad (e)$$

and

$$\theta'_{LF/A} = \frac{\theta'_L - \theta'_A}{\theta'_F - \theta'_A}, \quad (7.26)$$

where

θ'_F = kinematic coefficient of the first gear in the train,
 θ'_L = kinematic coefficient of the last gear in the train, and
 θ'_A = kinematic coefficient of the arm.

The following examples help to illustrate the use of kinematic coefficients presented in Eqs. (7.25) and (7.26).

EXAMPLE 7.4

Figure 7.36 shows a reverted planetary gear train. Gear 2 is fastened to its shaft and is driven at 250 rev/min in a clockwise direction. Gears 4 and 5 are planet gears that are joined but are free to turn on the shaft carried by the arm. Gear 6 is stationary. Find the speed and direction of rotation of the arm.

SOLUTION

We must first decide which gears to designate the first and last members of the train. Since the speeds of gears 2 and 6 are both given, then either gear may be chosen as the first. The choice makes no difference to the results, but once the decision is made, it may not be changed. Here we choose gear 2 as first; therefore, gear 6 is last. Thus, choosing counterclockwise as positive gives

$$\omega_F = \omega_2 = -250 \text{ rev/min} \quad \text{and} \quad \omega_L = \omega_6 = 0 \text{ rev/min},$$

and, according to Eq. (7.24), the first-order kinematic coefficient is

$$\theta'_{LF} = \theta'_{62} = \left(-\frac{16}{34}\right) \left(-\frac{20}{30}\right) = \frac{16}{51},$$

where the positive sign results, since there are two external contacts.

Substituting the values into Eq. (7.25) gives the first-order kinematic coefficient:

$$\theta'_{LF/A} = \frac{16}{51} = \frac{0 - \omega_3}{-250 \text{ rev/min} - \omega_3}.$$

Rearranging this equation gives the angular velocity of arm A:

$$\omega_A = \omega_3 = 114.3 \text{ rev/min ccw.} \quad \text{Ans.}$$

EXAMPLE 7.5

Consider the simple planetary gear train shown in Fig. 7.40 in which the ring gear h is fixed (that is, the angular velocity of the ring gear is $\omega_h = 0$) and the input is the planet carrier (the arm). The dimensions of ring gear h , planet gear j , sun gear k , and arm A are $R_h = 200 \text{ mm}$, $R_j = 50 \text{ mm}$, $R_k = 100 \text{ mm}$, and $R_A = 150 \text{ mm}$, respectively. If the input angular velocity is $\omega_A = \omega_i = 10 \text{ rad/s ccw}$, determine the angular velocities of the planet gear, j , and the sun gear, k .

SOLUTION

Using Eq. (7.25), the first-order kinematic coefficient of planet gear j , in rolling contact with ring gear h , can be written as

$$\theta'_{jh/A} = \pm \frac{R_h}{R_j} = \frac{\omega_j - \omega_A}{\omega_h - \omega_A}. \quad (1a)$$

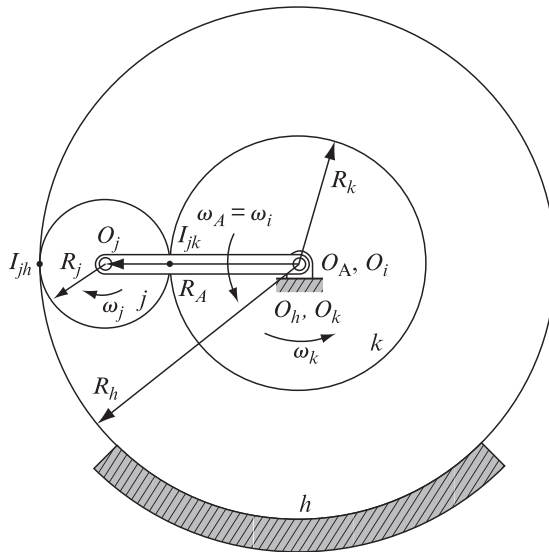


Figure 7.40 A simple planetary gear train with the ring gear fixed.

Similarly, the first-order kinematic coefficient of planet gear j , in rolling contact with sun gear k , can be written as

$$\theta'_{jk/A} = \pm \frac{R_k}{R_j} = \frac{\omega_j - \omega_A}{\omega_k - \omega_A}. \quad (1b)$$

Note that if the ring gear is removed, and the arm is fixed (that is, $\omega_A = 0$), then the planetary gear train reduces to an ordinary gear train, that is, Eq. (1b) reduces to

$$\theta'_{jk/A} = \pm \frac{R_k}{R_j} = \frac{\omega_j}{\omega_k}. \quad (2)$$

Since there is internal contact between the planet gear and the fixed ring gear, then the positive sign is used in Eq. (1a). Then, rearranging the equation and substituting $\omega_h = 0$ gives

$$R_j \omega_j = -(R_h - R_j) \omega_A = -R_A \omega_A. \quad (3)$$

Therefore, the angular velocity of the planet gear is

$$\omega_j = -\frac{R_A \omega_A}{R_j} = -\frac{(150 \text{ mm})(10 \text{ rad/s})}{50 \text{ mm}} = -30 \text{ rad/s.} \quad \text{Ans.}$$

The negative sign indicates that the direction of the angular velocity of the planet gear is opposite to the direction of the arm and, therefore, clockwise.

Similarly, since there is external contact between the planet gear and the sun gear, the negative sign is used in Eq. (1b), and the equation can be written as

$$R_k \omega_k = -R_j \omega_j + (R_k + R_j) \omega_A. \quad (4a)$$

The angular velocity of the sun gear can be expressed in terms of the input angular velocity of the arm by substituting Eq. (3) into Eq. (4a) and simplifying; that is,

$$R_k \omega_k = (R_k + R_h) \omega_A = 2R_A \omega_A. \quad (4b)$$

Therefore, the angular velocity of the sun gear is

$$\omega_k = \frac{2R_A \omega_A}{R_k} = \frac{2(150 \text{ mm})(10 \text{ rad/s})}{100 \text{ mm}} = 30 \text{ rad/s.} \quad \text{Ans.}$$

The positive result indicates that the direction of the angular velocity of the sun gear is the same as the arm and is, therefore, counterclockwise.

EXAMPLE 7.6

Consider the compound planetary gear train that is commonly used in an electric drill or screwdriver, shown in the schematic diagram in Fig. 7.41. The gear and the arm dimensions are $R_1 = 150 \text{ mm}$, $R_{2A} = R_{4A} = 100 \text{ mm}$, and $R_3 = R_4 = R_5 = R_6 = 50 \text{ mm}$.

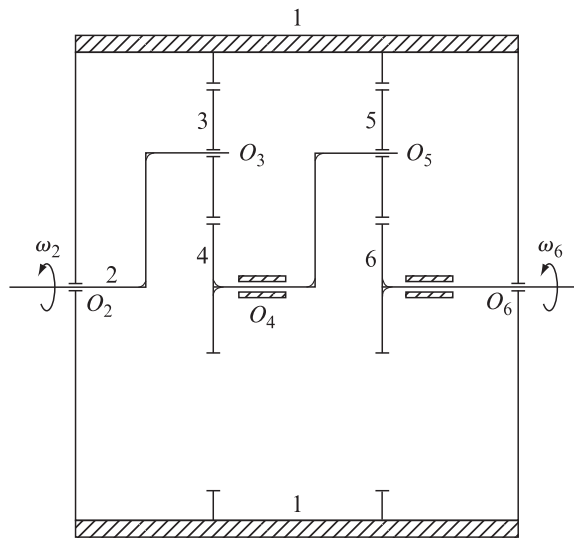


Figure 7.41 Schematic diagram of a compound planetary gear train.

The angular velocity of the input shaft (and arm 2) is $\omega_{2A} = 10$ rad/s ccw (looking from the left), and ring gear 1 is fixed—that is, the angular velocity of the ring gear is $\omega_1 = 0$. Determine the angular velocities of gears 3, 4, 5, and 6.

SOLUTION

This gear train is composed of two simple planetary gear trains in series; that is, the first planetary gear train is composed of ring gear 1, input arm 2, planet gear 3, and sun gear 4, and the second planetary gear train is composed of ring gear 1, arm 4, planet gear 5, and output sun gear 6. Note that the angular velocities of gear 4 and arm 4 are equal in magnitude and direction, since they are rigidly attached. Also, note that gear 4 acts as both the output of the first planetary gear train and the input to the second planetary gear train. Using Eq. (7.25), the first-order kinematic coefficient of gear 4 with respect to ring gear 1 can be written as

$$\theta'_{41/2A} = \left(-\frac{R_3}{R_4} \right) \left(\frac{R_1}{R_3} \right) = -\frac{R_1}{R_4} = \frac{\omega_4 - \omega_{2A}}{\omega_1 - \omega_{2A}}. \quad (1a)$$

The correct sign is negative, since gears 4 and 3 are in external contact, and gears 3 and 1 are in internal contact. Substituting the given data into Eq. (1a) gives

$$-\frac{150 \text{ mm}}{50 \text{ mm}} = \frac{\omega_4 - 10 \text{ rad/s}}{0 - 10 \text{ rad/s}}. \quad (1b)$$

Therefore, the angular velocity of gear 4 is

$$\omega_4 = 40 \text{ rad/s}, \quad \text{Ans.}$$

where the positive result implies that gear 4 is rotating in the same direction as the input gear 2—that is, counterclockwise.

Using Eq. (7.25), the first-order kinematic coefficient of gear 3 with respect to gear 4 can be written as

$$\theta'_{34/2A} = -\frac{R_4}{R_3} = \frac{\omega_3 - \omega_{2A}}{\omega_4 - \omega_{2A}}, \quad (2a)$$

where the negative sign denotes external contact between gears 3 and 4. Therefore, the first-order kinematic coefficient is

$$\theta'_{34/2A} = -\frac{50 \text{ mm}}{50 \text{ mm}} = \frac{\omega_3 - 10 \text{ rad/s}}{40 \text{ rad/s} - 10 \text{ rad/s}}. \quad (2b)$$

Therefore, the angular velocity of gear 3 is

$$\omega_3 = -20 \text{ rad/s}, \quad \text{Ans.}$$

where the negative sign indicates that gear 3 is rotating in the direction opposite to the input gear 2—that is, clockwise.

Using Eq. (7.25), the first-order kinematic coefficient of gear 6 with respect to the ring gear 1 can be written as

$$\theta'_{61/4A} = \left(-\frac{R_5}{R_6}\right) \left(\frac{R_1}{R_5}\right) = -\frac{R_1}{R_6} = \frac{\omega_6 - \omega_{4A}}{\omega_1 - \omega_{4A}}. \quad (3a)$$

The correct sign is negative, since gears 6 and 5 are in external contact, and gears 5 and 1 are in internal contact. Substituting the given data into Eq. (3a) gives

$$-\frac{150 \text{ mm}}{50 \text{ mm}} = \frac{\omega_6 - 40 \text{ rad/s}}{0 - 40 \text{ rad/s}}. \quad (3b)$$

Therefore, the angular velocity of gear 6 is

$$\omega_6 = 160 \text{ rad/s}, \quad \text{Ans.}$$

where the positive result implies that gear 4 is rotating in the same direction as the input gear 2—that is, counterclockwise.

Using Eq. (7.25), the first-order kinematic coefficient of gear 5 with respect to gear 6 can be written as

$$\theta'_{56/4A} = -\frac{R_6}{R_5} = \frac{\omega_5 - \omega_{4A}}{\omega_6 - \omega_{4A}}, \quad (4a)$$

where the negative sign denotes external contact between gears 5 and 6. Substituting the known data, the first-order kinematic coefficient is

$$\theta'_{56/4A} = -\frac{50 \text{ mm}}{50 \text{ mm}} = \frac{\omega_5 - 40 \text{ rad/s}}{160 \text{ rad/s} - 40 \text{ rad/s}}. \quad (4b)$$

Therefore, the angular velocity of gear 5 is

$$\omega_5 = -80 \text{ rad/s}, \quad \text{Ans.}$$

where the negative sign implies that gear 5 is rotating in the direction opposite to the input gear 2—that is, clockwise.

ALTERNATE SOLUTION

The rolling contact constraint between planet gear 3 and ring gear 1 can be written in terms of kinematic coefficients as

$$\frac{R_3}{R_1} = \frac{50 \text{ mm}}{150 \text{ mm}} = \frac{1}{3} = \frac{\theta'_{12} - \theta'_{22}}{\theta'_{32} - \theta'_{22}}, \quad (1)$$

where the positive sign is chosen, since planet gear 3 has internal contact with ring gear 1.

Since ring gear 1 is fixed, the kinematic coefficient of the ring gear is $\theta'_{12} = 0$ and, by definition, the kinematic coefficient of input arm 2 is $\theta'_{22} = 1$. Substituting these into Eq. (1), the kinematic coefficient of planet gear 3 with respect to input gear 2 is

$$\theta'_{32} = -2 \text{ rad/rad}, \quad (2)$$

where the negative sign indicates that gear 3 is rotating in the direction opposite to input gear 2—that is, clockwise. Therefore, the angular velocity of gear 3 is

$$\omega_3 = \theta'_{32}\omega_2 = -20 \text{ rad/s.} \quad \text{Ans.}$$

The rolling contact constraint between sun gear 4 and planet gear 3 can be written from Eq. (1) as

$$-\frac{R_4}{R_3} = -1 = \frac{\theta'_{32} - \theta'_{22}}{\theta'_{42} - \theta'_{22}}, \quad (3)$$

where the negative sign is chosen, since planet gear 3 has external contact with sun gear 4.

Substituting Eq. (2) and $\theta'_{22} = 1$, the known geometry, into Eq. (3), the kinematic coefficient of sun gear 4 with respect to gear 2 is

$$\theta'_{42} = 4 \text{ rad/rad}, \quad (4)$$

where the positive result implies that gear 4 is rotating in the same direction as input gear 2—that is, counterclockwise. Therefore, the angular velocity of gear 4 is

$$\omega_4 = \theta'_{42}\omega_2 = 40 \text{ rad/s ccw.} \quad \text{Ans.}$$

The rolling contact constraint between the planet gear 5 and ring gear 1 can be written from Eq. (1) as

$$\frac{R_5}{R_1} = \frac{50 \text{ mm}}{150 \text{ mm}} = \frac{1}{3} = \frac{\theta'_{12} - \theta'_{42}}{\theta'_{52} - \theta'_{42}}, \quad (5)$$

where the positive sign is used, since planet gear 5 has internal contact with ring gear 1.

Substituting Eq. (4) and $\theta'_{12} = 0$, the kinematic coefficient of planet gear 5 with respect to gear 2 is

$$\theta'_{52} = -8 \text{ rad/rad}, \quad (6)$$

where the negative sign indicates that gear 5 is rotating in the direction opposite to the input gear 2—that is, clockwise. Therefore, the angular velocity of gear 5 is

$$\omega_5 = \theta'_{52}\omega_2 = -80 \text{ rad/s (cw)}. \quad \text{Ans.}$$

The rolling contact constraint between the sun gear 6 and planet gear 5 can be written from Eq. (5) as

$$-\frac{R_6}{R_5} = -1 = \frac{\theta'_{52} - \theta'_{42}}{\theta'_{62} - \theta'_{42}}, \quad (7)$$

where the negative sign is used because of the external contact between planet gear 5 and sun gear 6.

Substituting Eqs. (4) and (6) and the known geometry into Eq. (7), the kinematic coefficient of sun gear 6 with respect to gear 2 is

$$\theta'_{62} = 16 \text{ rad/rad}, \quad (8)$$

where the positive result implies that gear 6 is rotating in the same direction as input gear 2. Therefore, the angular velocity of gear 6 is

$$\omega_6 = \theta'_{62}\omega_2 = 160 \text{ rad/s ccw}. \quad \text{Ans.}$$

EXAMPLE 7.7

Consider the planetary gear train shown in the schematic diagram in Fig. 7.42. The radius of the fixed sun gear is $R_1 = 100 \text{ mm}$, the radius of input gear 2 is $R_2 = 100 \text{ mm}$, the radius of ring gear 4 is $R_4 = 200 \text{ mm}$, and the radii of gears 3 and 5 are $R_3 = R_5 = 50 \text{ mm}$. The angular velocity of input shaft 2 is $\omega_2 = 10 \text{ rad/s ccw}$ (looking from the left). Determine the angular velocities of gears 3, and 4, and 5.

SOLUTION

The rolling contact constraint between input gear 2 and gear 5 can be written as

$$-\frac{R_2}{R_5} = -\frac{100 \text{ mm}}{50 \text{ mm}} = -2 = \frac{\theta'_{52}}{\theta'_{22}}. \quad (1)$$

Since the kinematic coefficient of input shaft 2 is $\theta'_{22} = 1 \text{ rad/rad}$, the kinematic coefficient of gear 5 is

$$\theta'_{52} = -2 \text{ rad/rad}, \quad (2)$$

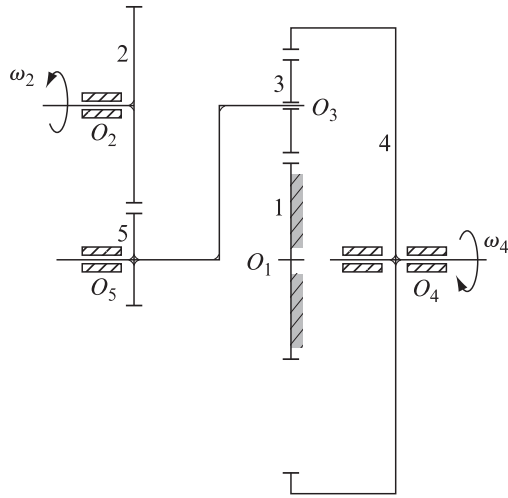


Figure 7.42 Schematic diagram of a planetary gear train.

where the negative sign implies that gear 5 is rotating in the opposite direction to input gear 2—that is, clockwise. Therefore, the angular velocity of gear 5 is

$$\omega_5 = \theta'_{52} \omega_2 = -20 \text{ rad/s (cw).} \quad \text{Ans.}$$

The rolling contact constraint between planet gear 3 and fixed sun gear 1 can be written as

$$-\frac{R_3}{R_1} = \frac{\theta'_{12} - \theta'_{5A2}}{\theta'_{32} - \theta'_{5A2}}, \quad (3)$$

which, for the given conditions, becomes

$$-\frac{50 \text{ mm}}{100 \text{ mm}} = \frac{0 + 2 \text{ rad/rad}}{\theta'_{32} + 2 \text{ rad/rad}}.$$

Therefore, the kinematic coefficient of gear 3 with respect to fixed sun gear 1 is

$$\theta'_{32} = -6 \text{ rad/rad} \quad (4)$$

and the angular velocity of gear 3 is

$$\omega_3 = \theta'_{32} \omega_2 = -60 \text{ rad/s,} \quad \text{Ans.}$$

where the negative sign indicates that the direction of ω_3 is clockwise.

The rolling contact constraint between ring gear 4 and planet gear 3 can be written as

$$\frac{R_3}{R_4} = \frac{\theta'_{42} - \theta'_{5A2}}{\theta'_{32} - \theta'_{5A2}}, \quad (5)$$

which, using Eqs. (2) and (4) and the given dimensions, can be written as

$$\frac{50 \text{ mm}}{200 \text{ mm}} = \frac{\theta'_{42} + 2 \text{ rad/rad}}{-6 \text{ rad/rad} + 2 \text{ rad/rad}}.$$

Therefore, the kinematic coefficient of gear 4 with respect to fixed sun gear 1 is

$$\theta'_{42} = -3 \text{ rad/rad} \quad (6)$$

and the angular velocity of gear 4 is

$$\omega_4 = \theta'_{42} \omega_2 = -30 \text{ rad/s}, \quad \text{Ans.}$$

where the negative sign indicates that the direction of ω_4 is clockwise.

The kinematic coefficients in Eqs. (2), (4), and (6) define the geometry of the gear train and provide the capability for obtaining the angular velocities (magnitudes and directions) in terms of the input angular velocity, ω_2 .

7.16 TABULAR ANALYSIS OF EPICYCLIC GEAR TRAINS

Another method of determining the rotational speeds of epicyclic gear trains uses the principle of superposition. The total analysis is carried out by finding the apparent rotations of the components with respect to the arm or planet carrier and then summing with the rotation of the components as if all the gears are fixed to the arm. The process is easily carried out in a tabular procedure.

Figure 7.35 shows a planetary gear train composed of sun gear 2, planet carrier (arm) 3, planet gear 4, and internal gear 5 that is in mesh with the planet gear. Since this gear train has two degrees of freedom, we might reasonably specify the angular velocities of both the sun gear and the arm, and wish to determine the angular velocity of the internal gear.

The analysis can be carried out in the following three steps:

1. Consider all gears (including the fixed gear, if any) to be locked to the arm, and allow the arm to rotate with angular velocity ω_A . Tabulate the angular velocities of all components under this condition as also equal to ω_A .
2. Free all constraints of step 1, fix the arm, and allow some other gear B (such as the sun gear, for example) to rotate with angular velocity $\omega_{B/A}$. Tabulate the apparent angular velocities of all other gears with respect to the arm as multiples of $\omega_{B/A}$.
3. Add the angular velocities of each gear from steps 1 and 2, and apply the given input velocities in order to find numeric values for ω_A and $\omega_{B/A}$.

A few examples will clarify how this can be done in a convenient tabular procedure.

EXAMPLE 7.8

Let us assume the tooth numbers shown in Fig. 7.35, and let the angular velocities of the sun gear and the arm be $\omega_2 = 100 \text{ rev/min}$ and $\omega_3 = 200 \text{ rev/min}$, respectively, both in the

counterclockwise direction, chosen positive. What is the angular velocity of internal ring gear 5?

SOLUTION

The solution process described by the three steps enumerated above is demonstrated in Table 7.3.

Table 7.3 Tabular Analysis for Examples 7.8 and 7.9

Step Number	Gear 2	Arm 3	Gear 4	Gear 5
1. Gears fixed to arm	ω_3	ω_3	ω_3	ω_3
2. Arm fixed	$\omega_{2/3}$	0	$(-40/20)\omega_{2/3}$	$(20/80)(-40/20)\omega_{2/3}$
3. Total	$\omega_3 + \omega_{2/3}$	ω_3	$\omega_3 + (-40/20)\omega_{2/3}$	$\omega_3 + (20/80)(-40/20)\omega_{2/3}$

Next, comparing the given input velocities with the bottom row of columns 2 and 3, we see that, with $\omega_2 = \omega_3 + \omega_{2/3} = 100$ rev/min and $\omega_3 = 200$ rev/min, then $\omega_{2/3} = -100$ rev/min. Therefore, from the bottom row of column 5, the angular velocity of internal ring gear 5 can be written as

$$\omega_5 = \omega_3 + (20/80)(-40/20)\omega_{2/3};$$

that is,

$$\omega_5 = (200 \text{ rev/min}) + (-40/80)(-100 \text{ rev/min}) = 250 \text{ rev/min},$$

where the positive result implies that gear 5 is rotating in the same direction as the positive inputs—that is, counterclockwise.

EXAMPLE 7.9

What is the angular velocity of internal gear 5 of Fig. 7.35 if gear 2 rotates at 100 rev/min cw while arm 3 rotates at 200 rev/min ccw?

SOLUTION

The analysis is identical to that performed in Table 7.3 for Example 7.8. However, the input velocities have now changed. Still taking counterclockwise as positive, the bottom row of columns 2 and 3 indicate that, since

$$\omega_2 = \omega_3 + \omega_{2/3} = -100 \text{ rev/min} \quad \text{and} \quad \omega_3 = 200 \text{ rev/min},$$

then $\omega_{2/3} = -300$ rev/min. Therefore, the bottom row of column 5 indicates

$$\begin{aligned}\omega_5 &= \omega_3 + (20/80)(-40/20)\omega_{2/3} = (200 \text{ rev/min}) + (-40/80)(-300 \text{ rev/min}) \\ &= 350 \text{ rev/min ccw.}\end{aligned}$$

Ans.

EXAMPLE 7.10

The planetary gear train shown in Fig. 7.43 is called *Ferguson's paradox*.^{*} Gear 2 is fixed to the frame. Arm 3 and gears 4, 5, and 6 are free to turn upon their shafts. Gears 2, 4, and 5 have tooth numbers of 100, 101, and 99, respectively, all cut with the same circular pitch (but with slightly different pitch circle radii) so that planet gear 6 meshes with all of them. Find the angular rotations of gears 4 and 5 when arm 3 is given one counterclockwise turn.

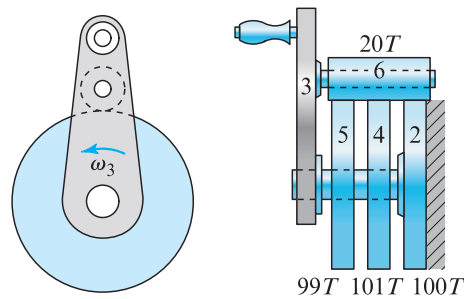


Figure 7.43 Ferguson's paradox.

SOLUTION

The solution process is indicated in Table 7.4.

Table 7.4 Tabular Analysis for Example 7.10

Step Number	Gear 2	Arm 3	Gear 4	Gear 5	Gear 6
1. Gears fixed to arm	$\Delta\theta_3$	$\Delta\theta_3$	$\Delta\theta_3$	$\Delta\theta_3$	$\Delta\theta_3$
2. Arm fixed	$\Delta\theta_{2/3}$	0	$(-20/101)(-100/20)\Delta\theta_{2/3}$	$(-20/99)(-100/20)\Delta\theta_{2/3}$	$(-100/20)\Delta\theta_{2/3}$
3. Total	$\Delta\theta_3 + \Delta\theta_{2/3}$	$\Delta\theta_3$	$\Delta\theta_3 + (-20/101)(-100/20)\Delta\theta_{2/3}$	$\Delta\theta_3 + (-20/99)(-100/20)\Delta\theta_{2/3}$	$\Delta\theta_3 + (-100/20)\Delta\theta_{2/3}$

It should be noted that angular displacements are indicated in Table 7.4 instead of angular velocities. This is caused by the nature of the question asked and recognizing that, during a chosen time interval, $\Delta\theta = \omega\Delta t$ for each of the elements. According to the problem statement, we choose the time interval Δt such that for the arm $\Delta\theta_3 = 1$

^{*}James Ferguson (1710–1776), Scottish physicist and astronomer, first published this device under the title *The Description and Use of a New Machine Called the Mechanical Paradox*, London, 1764.

rev ccw. Then, for gear 2 to remain stationary, column 2 indicates that $\Delta\theta_3 + \Delta\theta_{2/3} = (1 \text{ rev}) + \Delta\theta_{2/3} = 0$, and, therefore, $\Delta\theta_{2/3} = -1 \text{ rev}$. Finally, from the bottom row of columns 4 and 5, we determine that

$$\Delta\theta_4 = \Delta\theta_3 + (-20/101)(-100/20)\Delta\theta_{2/3} = (1 \text{ rev}) + (100/101)(-1 \text{ rev}) = 1/101 \text{ rev}$$

and

$$\Delta\theta_5 = \Delta\theta_3 + (-20/99)(-100/20)\Delta\theta_{2/3} = (1 \text{ rev}) + (100/99)(-1 \text{ rev}) = -1/99 \text{ rev}.$$

Thus, when arm 3 is turned 1 rev ccw, gear 4 rotates 1/101 rev ccw, whereas gear 5 turns 1/99 rev cw. *Ans.*

EXAMPLE 7.11

The overdrive unit shown in Fig. 7.44 is sometimes used following a standard automotive transmission to further reduce engine speed. The engine speed (after the transmission) corresponds to the speed of planet carrier 3, and the drive shaft speed corresponds to that of gear 5; sun gear 2 is held stationary. Determine the percentage additional reduction in engine speed obtained when the overdrive is active.

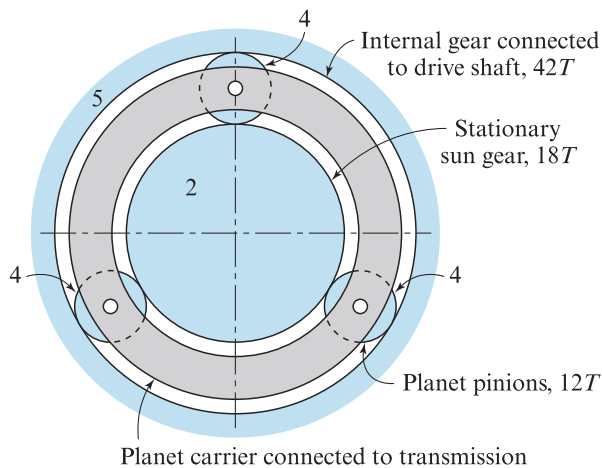


Figure 7.44 Overdrive unit.

SOLUTION

The analysis for this problem is demonstrated in Table 7.5. For gear 2 to remain stationary, column 2 indicates that $\omega_3 + \omega_{2/3} = 0$; therefore $\omega_{2/3} = -\omega_3$. Putting this into column 5 indicates that

$$\omega_5 = \omega_3 + (12/42)(-18/12)\omega_{2/3} = \omega_3 + (-18/42)(-\omega_3) = 1.429\omega_3.$$

Therefore, the percentage reduction in engine speed is

$$(1.429\omega_3 - 1.0\omega_3)/(1.429\omega_3) = 0.300 = 30\%.$$

Ans.

Table 7.5 Tabular Analysis for Example 7.11

Step Number	Gear 2	Arm 3	Gear 4	Gear 5
1. Gears fixed to arm	ω_3	ω_3	ω_3	ω_3
2. Arm fixed	$\omega_{2/3}$	0	$(-18/12)\omega_{2/3}$	$(12/42)(-18/12)\omega_{2/3}$
3. Total	$\omega_3 + \omega_{2/3}$	ω_3	$\omega_3 + (-18/12)\omega_{2/3}$	$\omega_3 + (12/42)(-18/12)\omega_{2/3}$

7.17 REFERENCES

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- [3] Pennock, G. R., and J. J. Alwerdt, 2007. Duality between the kinematics of gear trains and the statics of beam systems, *Mech. Mach. Theory* **42**(11):512–26.
- [4] Shigley, J. E., and C. R. Mischke, 2001. *Mechanical Engineering Design*, 6th ed., Boston: McGraw-Hill.

PROBLEMS

- 7.1** Find the diametral pitch of a pair of gears having 32 and 84 teeth, respectively, whose center distance is 3.625 in.
- 7.2** Find the number of teeth and the circular pitch of a 6-in pitch-diameter gear whose diametral pitch is 9 teeth/in.
- 7.3** Determine the module of a pair of gears having 18 and 40 teeth, respectively, whose center distance is 58 mm.
- 7.4** Find the number of teeth and the circular pitch of a gear whose pitch diameter is 200 mm if the module is 8 mm/tooth.
- 7.5** Find the diametral pitch and the pitch diameter of a 40-tooth gear whose circular pitch is 3.50 in/tooth.
- 7.6** The pitch diameters of a pair of mating gears are 3.50 in and 8.25 in, respectively. If the diametral pitch is 16 teeth/in, how many teeth are there on each gear?
- 7.7** Find the module and the pitch diameter of a gear whose circular pitch is 40 mm/tooth if the gear has 36 teeth.
- 7.8** The pitch diameters of a pair of gears are 60 mm and 100 mm, respectively. If their module is 2.5 mm/tooth, how many teeth are there on each gear?
- 7.9** What is the diameter of a 33-tooth gear if its circular pitch is 0.875 in/tooth?
- 7.10** A shaft carries a 30-tooth, 3-teeth/in diametral pitch gear that drives another gear at a speed of 480 rev/min. How fast does the 30-tooth gear rotate if the shaft center distance is 9 in?
- 7.11** Two gears having an angular velocity ratio of 3:1 are mounted on shafts whose centers are 136 mm apart. If the module of the gears is 4 mm/tooth, how many teeth are there on each gear?
- 7.12** A gear having a module of 4 mm/tooth and 21 teeth drives another gear at a speed of 240 rev/min. How fast is the 21-tooth gear rotating if the shaft center distance is 156 mm?
- 7.13** A 4-tooth/in diametral pitch, 24-tooth pinion is to drive a 36-tooth gear. The gears are cut on the 20° full-depth involute system. Find and tabulate the addendum, dedendum, clearance, circular pitch, base pitch, tooth thickness, pitch circle radii, base circle radii, lengths of paths of approach and recess, and contact ratio.
- 7.14** A 5-tooth/in diametral pitch, 15-tooth pinion is to mate with a 30-tooth internal gear. The gears are 20° full-depth involute. Make a drawing of the gears showing several teeth on each gear. Can these gears be assembled in a radial direction? If not, what remedy should be used?

- 7.15** A $2\frac{1}{2}$ -teeth/in diametral pitch 17-tooth pinion and a 50-tooth gear are paired. The gears are cut on the 20° full-depth involute system. Find the angles of approach and recess of each gear, and the contact ratio.
- 7.16** A gearset with a module of 5 mm/tooth has involute teeth with $22\frac{1}{2}^\circ$ pressure angle, and 19 and 31 teeth, respectively. They have 1.0 m for the addendum and 1.25 m for the dedendum.* Tabulate the addendum, dedendum, clearance, circular pitch, base pitch, tooth thickness, base circle radii, and contact ratio.
- 7.17** A gear with a module of 8 mm/tooth and 22 teeth is in mesh with a rack; the pressure angle is 25° . The addendum and dedendum are 1.0 m and 1.25 m, respectively.* Find the lengths of the paths of approach and recess, and determine the contact ratio.
- 7.18** Repeat Problem 7.15 using the 25° full-depth system.
- 7.19** Draw a 2-tooth/in diametral pitch, 26-tooth, 20° full-depth involute gear in mesh with a rack. (a) Find the lengths of the paths of approach and recess, and the contact ratio. (b) Draw a second rack in mesh with the same gear but offset $1/8$ in farther away from the gear center. Determine the new contact ratio. Has the pressure angle changed?
- 7.20 to 7.24** Shaper gear cutters have the advantage that they can be used for either external or internal gears and also that only a small amount of runout is necessary at the end of the stroke. The generating action of a pinion shaper cutter can easily be simulated by employing a sheet of clear plastic. Figure P 7.20 shows 1 tooth of a 16-tooth pinion cutter with 20° pressure angle as it can be cut

from a plastic sheet. To construct the cutter, lay out the tooth on a sheet of drawing paper. Be sure to include the clearance at the top of the tooth. Draw radial lines through the pitch circle spaced at distances equal to one fourth of the tooth thickness, as shown in the figure. Next, fasten the sheet of plastic to the drawing and scribe the cutout, the pitch circle, and the radial lines onto the sheet. Then, remove the sheet and trim the tooth outline with a razor blade. Next, use a small piece of fine sandpaper to remove any burrs.

To generate a gear with the cutter, only the pitch circle and the addendum circle need be drawn. Divide the pitch circle into spaces equal to those used on the template and construct radial lines through them. The tooth outlines are then obtained by rolling the template pitch circle upon that of the gear and drawing the cutter tooth lightly for each position. The resulting generated tooth upon the gear will be evident. The following problems all employ a standard 1-tooth/in diametral pitch 20° full-depth template constructed as described earlier. In each, case, you should generate a few teeth and estimate the amount of undercutting.

Table P7.20 to P7.24

Problem no.	P7.20	P7.21	P7.22	P7.23	P7.24
No. of teeth	10	12	14	20	36

- 7.25** A 10-mm/tooth module gear has 17 teeth, a 20° pressure angle, an addendum of 1.0 m, and a dedendum of 1.25 m.* Find the thickness of the teeth at the base circle and at the addendum circle. What is the pressure angle corresponding to the addendum circle?

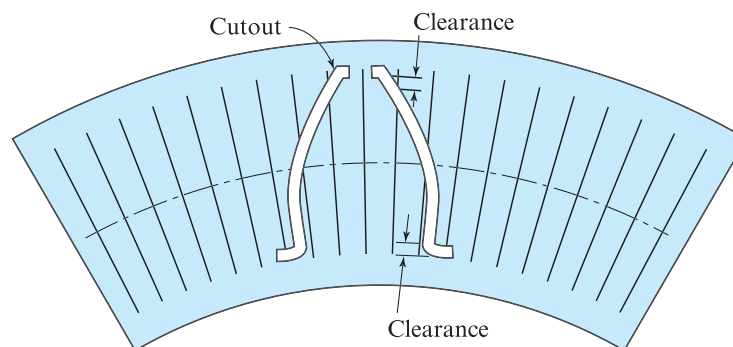


Figure P7.20

* In SI, tooth sizes are given in modules, m, and $a = 1.0$ m means 1 module, not 1 meter.

- 7.26** A 15-tooth pinion has $1\frac{1}{2}$ -tooth/in diametral pitch, 20° full-depth involute teeth. Calculate the thickness of the teeth at the base circle. What are the tooth thickness and the pressure angle at the addendum circle?
- 7.27** A tooth is 0.785 in thick at a pitch circle radius of 8 in and has a pressure angle of 25° . What is the thickness at the base circle?
- 7.28** A tooth is 1.571 in thick at the pitch radius of 16 in and has a pressure angle of 20° . At what radius does the tooth become pointed?
- 7.29** A 25° full-depth involute, 12-tooth/in diametral pitch pinion has 18 teeth. Calculate the tooth thickness at the base circle. What are the tooth thickness and pressure angle at the addendum circle?
- 7.30** A nonstandard 10-tooth 8-tooth/in diametral pitch involute pinion is to be cut with a $22\frac{1}{2}^\circ$ pressure angle. What maximum addendum can be used before the teeth become pointed?
- 7.31** The accuracy of cutting gear teeth can be measured by fitting hardened and ground pins in diametrically opposite tooth spaces and measuring the distance over the pins. For a 10-tooth/in diametral pitch, 20° full-depth involute system 96-tooth gear:
- Calculate the pin diameter that will contact the teeth at the pitch lines if there is to be no backlash.
 - What should be the distance measured over the pins if the gears are cut accurately?
- 7.32** A set of interchangeable gears with 4-tooth/in diametral pitch is cut on the 20° full-depth involute system. The gears have tooth numbers of 24, 32, 48, and 96. For each gear, calculate the radius of curvature of the tooth profile at the pitch circle and at the addendum circle.
- 7.33** Calculate the contact ratio of a 17-tooth pinion that drives a 73-tooth gear. The gears are 96-tooth/in diametral pitch and cut on the 20° full-depth involute system.
- 7.34** A 25° pressure angle 11-tooth pinion is to drive a 23-tooth gear. The gears have a diametral pitch of 8 teeth/in and have involute stub teeth. What is the contact ratio?
- 7.35** A 22-tooth pinion mates with a 42-tooth gear. The gears have full-depth involute teeth, have a diametral pitch of 16 teeth/in, and are cut with a $17\frac{1}{2}^\circ$ pressure angle.* Find the contact ratio.
- 7.36** The center distance of two 24-tooth, 20° pressure angle, full-depth involute spur gears with diametral pitch of 2 teeth/in is increased by 0.125 in over the standard distance. At what pressure angle do the gears operate?
- 7.37** The center distance of two 18-tooth, 25° pressure angle, full-depth involute spur gears with diametral pitch of 3 teeth/in is increased by 0.062 5 in over the standard distance. At what pressure angle do the gears operate?
- 7.38** A pair of mating gears have 24 teeth/in diametral pitch and are generated on the 20° full-depth involute system. If the tooth numbers are 15 and 50, what maximum addendums may they have if interference is not to occur?
- 7.39** A set of gears is cut with a $4\frac{1}{2}$ -in/tooth circular pitch and a $17\frac{1}{2}^\circ$ pressure angle.* The pinion has 20 full-depth teeth. If the gear has 240 teeth, what maximum addendum may it have to avoid interference?
- 7.40** Using the method described for Problems 7.20 to 7.24, cut a 1-tooth/in diametral pitch, 20° pressure angle full-depth involute rack tooth from a sheet of clear plastic. Use a nonstandard clearance of $0.35/P$ in order to obtain a stronger fillet. This template can be used to simulate the generating action of a hob. Now, using the variable-center-distance system, generate an 11-tooth pinion to mesh with a 25-tooth gear without interference. Record the values found for center distance, pitch radii, pressure angle, gear blank diameters, cutter offset, and contact ratio. Note that more than one satisfactory solution exists.
- 7.41** Using the template cut in Problem 7.40, generate an 11-tooth pinion to mesh with a 44-tooth gear with the long-and-short-addendum system. Determine and record suitable values for gear and pinion addendum and dedendum, and for the cutter offset and contact ratio. Compare the contact ratio with that of standard gears.
- 7.42** A pair of involute spur gears with 9 and 36 teeth are to be cut with a 20° full-depth cutter with diametral pitch of 3 teeth/in.
- Determine the amount that the addendum of the gear must be decreased to avoid interference.
 - If the addendum of the pinion is increased by the same amount, determine the contact ratio.

* Such gears came from an older standard and are now obsolete.

- 7.43** A standard 20° pressure angle full-depth involute 1-tooth/in diametral pitch 20-tooth pinion drives a 48-tooth gear. The speed of the pinion is 500 rev/min. Using the position of the point of contact along the line of action as the abscissa, plot a curve indicating the sliding velocity at all points of contact. Note that the sliding velocity changes sign when the point of contact passes through the pitch point.

- 7.44** Find the first-order kinematic coefficient of the gear train. What are the speed and direction of rotation of gear 8?

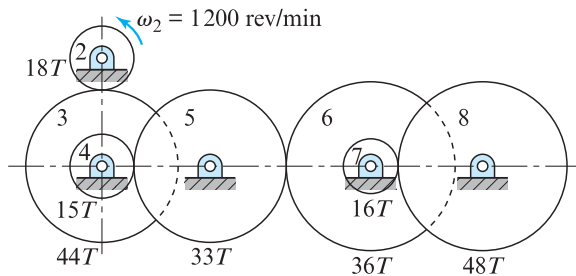


Figure P7.44

- 7.45** For the given pitch diameters of a set of spur gears forming a train, compute the first-order kinematic coefficient of the train. Determine the speeds and directions of rotation of gears 5 and 7.

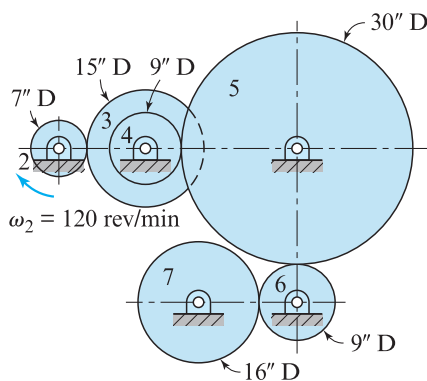


Figure P7.45

- 7.46** Use the truck transmission of Fig. 7.30 and an input speed of 3 000 rev/min to find the drive shaft speed for each forward gear and for the reverse gear.
- 7.47** Consider the gears in a speed-change gearbox used in machine tool applications. By sliding the cluster gears on shafts *B* and *C*, nine speed changes can be obtained. The problem of the machine tool designer is to select tooth numbers for the various gears

to produce a reasonable distribution of speeds for the output shaft. The smallest and largest gears are gears 2 and 9, respectively. Using 20 and 45 teeth for these gears, determine a set of suitable tooth numbers for the remaining gears. What are the corresponding speeds of the output shaft? Note that the problem has many solutions.

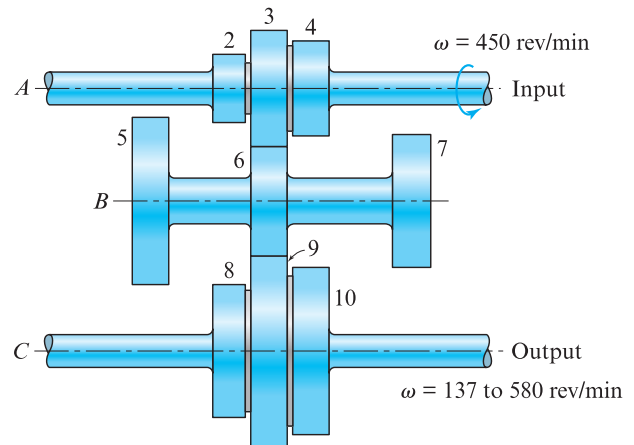


Figure P7.47

- 7.48** If internal gear 7 rotates at 60 rev/min ccw, determine the speed and direction of rotation of arm 3.

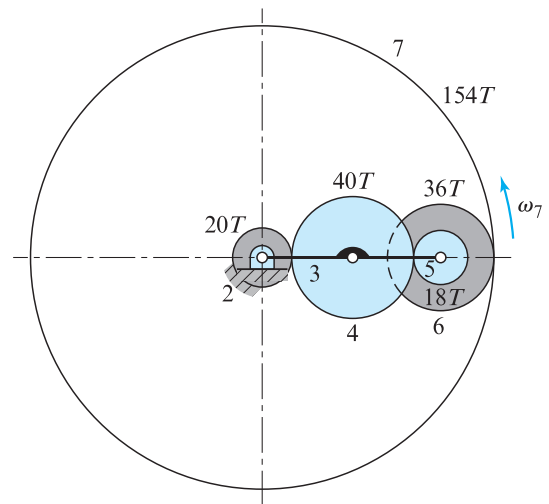


Figure P7.48

- 7.49** If the arm in Fig. P7.48 rotates at 300 rev/min ccw, find the speed and direction of rotation of internal gear 7.
- 7.50** If shaft *C* is stationary and gear 2 rotates at 800 rev/min ccw, determine the speed and direction of rotation of shaft *B*.

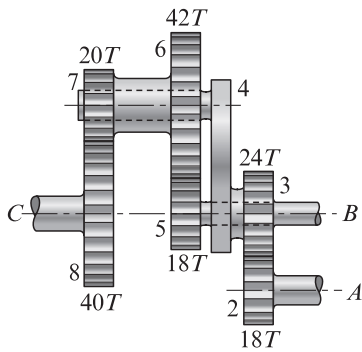


Figure P7.50

- 7.51** In Fig. P7.50, shaft B is stationary and shaft C is driven at 380 rev/min ccw. Determine the speed and direction of rotation of shaft A .
- 7.52** In Fig. P7.50, determine the speed and direction of rotation of shaft C if: (a) shafts A and B both rotate at 360 rev/min ccw; and (b) shaft A rotates at 360 rev/min cw and shaft B rotates at 360 rev/min ccw.
- 7.53** Gear 2 is connected to the input shaft, and arm 3 is connected to the output shaft. Determine the speed reduction. What is the sense of rotation of the output shaft? What changes could be made in the train to produce the opposite sense of rotation for the output shaft?

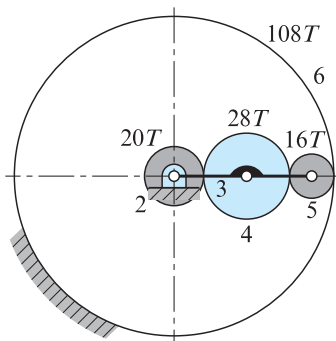


Figure P7.53

- 7.54** The Lévi type-L train shown in Fig. 7.38 has $N_2 = 16T$, $N_4 = 19T$, $N_5 = 17T$, $N_6 = 24T$, and $N_7 = 95T$. Internal gear 7 is fixed. Find the speed and direction of rotation of the arm if gear 2 is driven at 100 rev/min cw.
- 7.55** The Lévi type-A train of Fig. 7.38 has $N_2 = 20T$ and $N_4 = 32T$.
- (a) If the module is $m = 6$ mm/tooth, find the number of teeth on gear 5 and the crank arm radius.

- (b) If gear 2 is fixed and internal gear 5 rotates at 10 rev/min ccw, find the speed and direction of rotation of the arm.

7.56 The figure shows a possible arrangement of gears in a lathe headstock. Shaft A is driven by a motor at a speed of 720 rev/min. The three pinions can slide along shaft A to yield the meshes 2 with 5, 3 with 6, or 4 with 8. The gears on shaft C can also slide to mesh either 7 with 9 or 8 with 10. Shaft C is the mandrel shaft.

- (a) Make a table demonstrating all possible gear arrangements, beginning with the slowest speed for shaft C and ending with the highest, and enter in this table the speeds of shafts B and C .
- (b) If the gears all have a module of $m = 5$ mm/tooth, what must be the shaft center distances?

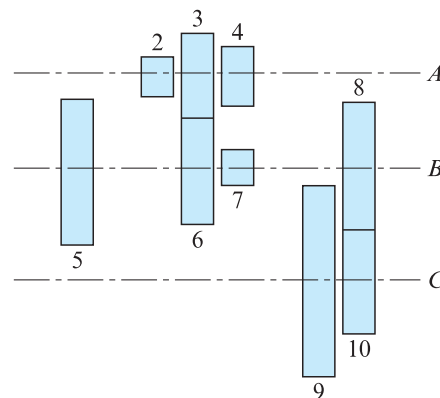


Figure P7.56 $N_2 = 16T$, $N_3 = 36T$, $N_4 = 25T$, $N_5 = 64T$, $N_6 = 66T$, $N_7 = 17T$, $N_8 = 55T$, $N_9 = 79T$, and $N_{10} = 41T$.

- 7.57** If shaft A is the output connected to the arm, and shaft B is the input driving gear 2, determine the speed ratio. Can you identify the Lévi type for this train?

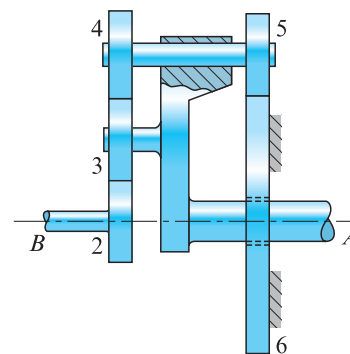


Figure P7.57 $N_2 = 16T$, $N_3 = 18T$, $N_4 = 16T$, $N_5 = 18T$, and $N_6 = 50T$.

- 7.58** In Problem 7.57, shaft B rotates at 100 rev/min cw. Find the speed of shaft A and of gears 3 and 4 about their own axes.
- 7.59** In the clock mechanism, a pendulum on shaft A drives an anchor (Fig. 1.12c). The pendulum period is such that 1 tooth of the 30T escapement wheel on shaft B is released every 2 s, causing shaft B to rotate once every minute. Note that the second (to the right) 64T gear is pivoted loosely on shaft D and is connected by a tubular shaft to the hour hand. (a) Show that the train values are such that the minute hand rotates once every hour and that the hour hand rotates once every 12 hours. (b) How many turns does the drum on shaft F make every day?

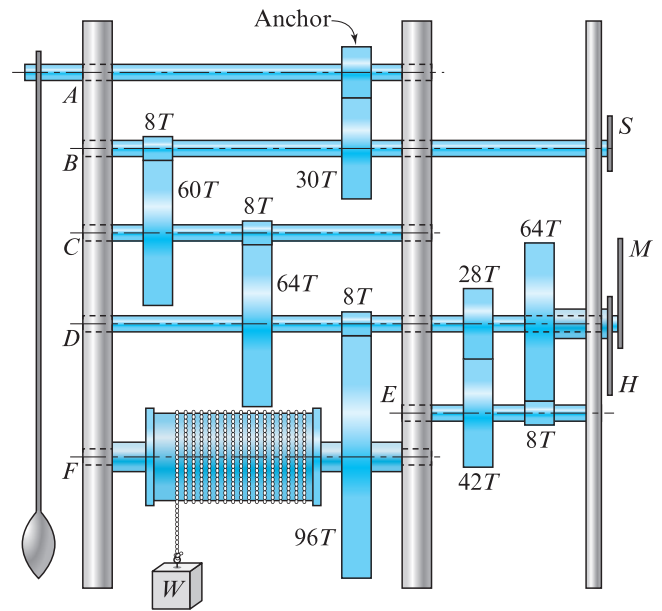


Figure P7.59 Clockwork mechanism.